

Cboe Futures Exchange Binary Order Entry Specification

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1 Introduction

1.1 Overview

This document describes Binary Order Entry (BOE), the CBOE proprietary order entry protocol used by a Trading Privilege Holder ("TPH") to connect to CBOE Futures Exchange ("CFE").

Where applicable, the terminology (e.g., time in force) used in this document is similar to that used by the FIX protocol to allow those familiar with FIX to more easily understand BOE. This document assumes the reader has basic knowledge of the FIX protocol.

BOE fulfills the following requirements:

- *CPU and memory efficiency.* Message encoding, decoding, and parsing are simpler to code and can be optimized to use less CPU and memory at runtime.
- *Application level simplicity.* State transitions are simple and unambiguous. They are easy to apply to a TPH's representation of an order.
- *Session level simplicity.* The session level protocol (login, sequencing, replay of missed messages, logout) is simple to understand.

While CFE has strived to preserve feature parity between FIX and BOE where possible, some features may only be available in one protocol or the other.

Note that while FIX is an ASCII based protocol, BOE is binary based providing for efficiencies that can allow for reduced latency.

All binary values are in little Endian (used by Intel x86 processors), and not network byte order.

Each message is identified by a unique message type. Not all message types are used in all of CFE's trading environments globally. A listing of the supported message types is provided in 'Section 10 - List of Message Types'.

All communication is via standard TCP/IP.

1.2 Hours of Operation

Trading hours on CFE vary by product. See the product contract specifications for details on trading hours for each product, which may differ for expiring and non-expiring contracts. The test product "ZVXT" will enter a queuing period five minutes ahead of other CFE products. TPHs are encouraged to utilize ZVXT for the test purposes of confirming connectivity and ensuring proper code path initialization during the five minutes prior to the start of the Pre-Open queuing period for other CFE products. See the Cboe Futures Exchange holiday calendar for trading hour adjustments corresponding to holidays.

BOE sessions are available for connection on Sunday starting at 10:30 a.m. CT. BOE sessions will disconnect each day between 4:05 and 4:45 p.m. CT for the daily restart. This will reset all sequences to zero in preparation for the next trading segment. BOE sessions will disconnect on Friday at around 4:05 p.m. CT but will remain available for connectivity testing (telnet testing) until startup on the following Sunday.

1.3 Data Types

The following data types are used by BOE. The size of some data types varies by message. All data types have default values of binary zero, in both TPH to CFE and CFE to TPH contexts.

- *Binary:* Little Endian byte order, unsigned binary value. The number of bytes used depends on the context.
  - One byte: `FE = 254`
  - Four bytes: `64 00 00 00 = 100`

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- *Signed Binary:* Little Endian byte order, signed two's complement, binary value. The number of bytes used depends on the context.
  - One byte: `DF = -33`
  - Four bytes: `64 00 00 00 = +100`
- *Binary Price:* Little Endian byte order value, signed two's complement, eight bytes in size, with four implied decimal places. So, if the value is −123,400, the actual value taking into account implied decimal places is −12.34.
  - `08 E2 01 00 00 00 00 00` = 123,400/10,000 = 12.34
  - `F8 1D FE FF FF FF FF FF` = −123,400/10,000 = −12.34
- *Alpha:* Uppercase letters (A-Z) and lowercase letters (a-z) only. ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- *Alphanumeric:* Uppercase letters (A-Z), lowercase letters (a-z) and numbers (0-9) only. ASCII NUL (0x00) filled on the right, if necessary.
- *Text:* Printable ASCII characters only. ASCII NUL (0x00) filled on the right, if necessary.
- *DateTime:* 8 bytes. The date and time, in UTC, represented as nanoseconds past the UNIX epoch (00:00:00 UTC on 1 January 1970). The nanoseconds portion is currently ignored and treated as 0 (i.e. the times are only accurate to microseconds) on input, and will always be set to 0 by CFE in outgoing messages. However, CFE may begin populating the nanoseconds portion at any time without warning.

  For example: 1,294,909,373,757,324,000 = 2011-01-13 09:02:53.757324 UTC.
- *Date:* Little Endian byte order, unsigned binary value, 4 bytes in size. The YYYYMMDD expressed as an integer.

1.4 Optional Fields and Bit fields

Some messages such as `New Order` and `Modify Order` have a number of optional fields. A count and number of bitfields in the message specify which optional fields will be present at the end of the message. If a bit is set, the field will be present. Fields are appended to the end of the message. There is no implicit framing between the optional fields. In order to decode the optional fields, they must be appended in a particular order to the end of the message. The fields of the first bitfield are appended first, lowest order bit first. Next, the fields of the next bit field are appended, lowest order bit first. This continues for all bitfields. While certain reserved bits within a defined bitfield are used within another CBOE market and will be ignored, bits that are reserved for future expansion must be set to 0 when noted in the bitfield description.

The size, data type, and values for each field are described in 'Section 8 – List of Optional Fields'.

Note that the set of optional fields returned for each CFE to TPH message type is determined at session login (using the `Login Request` message); hence, the exact size and layout of each message received by the client application can be known in advance. Any requested optional field which is irrelevant in a particular context will still be present in the returned message, but with all bytes set to binary zero (0x00).

Each return message from CFE to TPH indicates the optional fields which are present, even though the TPH indicated during login which optional fields are to be sent. The reason for the inclusion (and duplication) is so that each message can be interpreted on its own, without having to find the corresponding login request or response to know which optional fields are present. So, for example, in a log file, decoding a message requires only that single message.

Example messages are shown with each message type which should help to make this concept clear.

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1.5 Protocol Features

1.5.1 Architecture

Each BOE order handler process will allow a single TCP connection from a member. Connection attempts from unknown source IP ranges will be blocked to prevent unauthorized access to BOE ports. The Cboe NOC should be contacted in the event that a Member desires to connect from a new source IP range.

Each BOE order handler will connect, using a proprietary UDP protocol, to all matching units. Connections from order handlers to matching engines are latency equalized. The connections between order handlers and matching units are governed by an internal flow control mechanism to control burst rates.

1.5.2 Carried Order and Quote Restatements

Good 'till Cancel ("GTC") orders, Good 'till Date-Time ("GTD") orders, and Day orders or quotes entered during partial holiday sessions can result in orders persisting between sessions. The CFE BOE protocol provides a mechanism for clients to request restatement of orders that have been carried forward from the previous business day trading session. See 'Section 11 – Port Attributes' for information on available port attributes, including 'Carried Order Restatements'.

When enabled, Carried Order Restatements are sent to connected clients for each product on the CFE for which orders or quotes have been carried forward from the previous business day trading session. Carried Order Restatements are sent after connection establishment and before regular trading activity messages on a per-product basis.

Carried Order Restatements are represented using `Order Acknowledgement` messages with the following optional attributes set;

- *BaseLiquidityIndicator* = A (Added Liquidity), bitfield 5, bit position 7
- *SubLiquidityIndicator* = C (Carried), bitfield 7, bit position 1

To receive Carried Order Restatements, the Carried Order Restatement port attribute must be set (contact CFE Trade Desk). In addition, the following `Logon Request` message requirements must be met to receive Carried Order Restatements:

- Customers **must** register to receive *BaseLiquidityIndicator* and *SubLiquidityIndicator* optional fields on `Order Acknowledgement` messages via the `Logon Request` message (See 'Section 3.1.1 – Login Request' for details on registering to receive optional fields on a per-message basis). If the Carried Order Restatement port attribute is set and the bitfield Logon Message registration for the Order Acknowledgement message does not include but BaseLiquidityIndicator and SubLiquidityIndicator, the logon attempt will fail.
- Since the Carried Order Restatement messages are delivered to the session handler before the TPH connects, replay must be requested by setting the *NoUnspecifiedUnitReplay* parameter of the `Logon Request` message Unit Sequence Paramater Group to zero (i.e., don't suppress replay) or specifically set the UnitSequence to zero in the associated unit param group.

Note that no notification is provided at the end of a trading session to indicate when GTC, DTD, or Day orders/quotes on partial holiday sessions are persisted to carry over to the next trading sessions. Instead, Carried Order Restatements can be used by TPHs to be notified of orders/quotes that have persisted from the previous session.

1.5.2.1 Quotes Carried Across Multiple Sessions

Quotes cannot be marked as GTC or GTD, but Day quotes can persist across multiple sessions in the case of a holiday. The same Carried Order Restatement logic applies to quotes, which means that quotes will be restated with `Order Acknowledgement` messages containing the *Order ID* from the original `Quote Update Acknowledgement`. Quotes that are carried across multiple sessions may only be modified our cancelled by using a `Quote Update` message on the quote port where the quote originated. **TPHs may determine which restatements are quotes by**

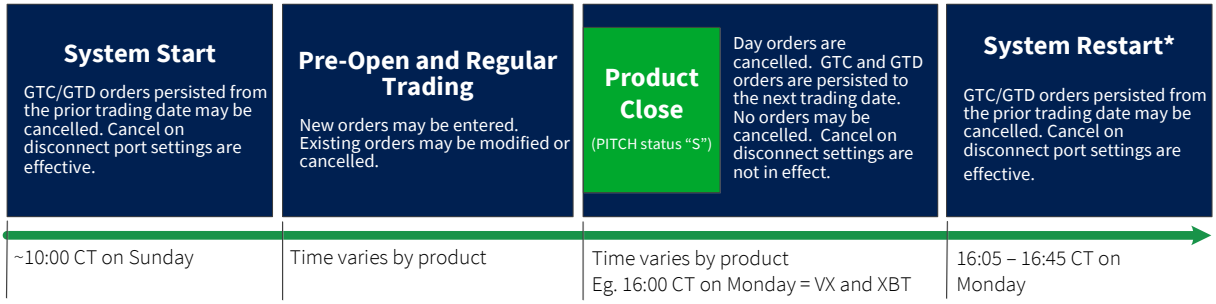
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matching the *OrderID* received on a carried order restatement (`Order Acknowledgement`) to the *OrderID* received in the original `Quote Update Acknowledgement`.

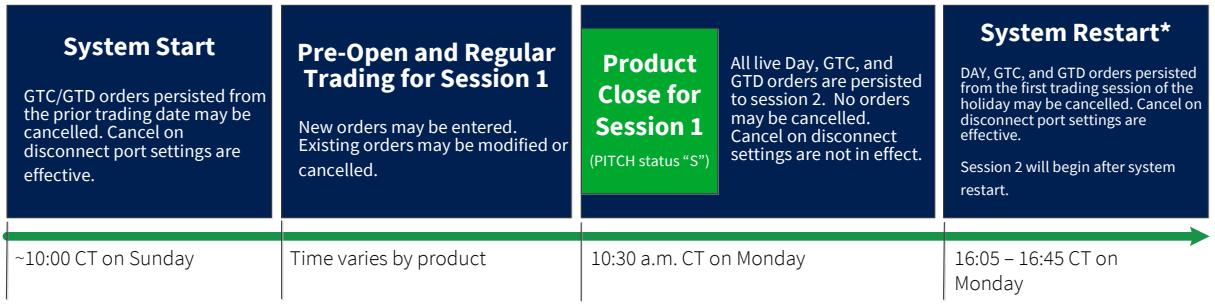
1.5.3 Cancellation of Carried Orders or Quotes Between Sessions

GTC and GTD orders persist within CFE's trading system between CFE business days. GTC, GTD, and Day orders/quotes also persist between multiple trading sessions on the same business day in connection with a holiday. Persisted orders/quotes can be cancelled while the associated product is in a suspended state and during other trading states as described above. At the scheduled end of trading for a product, cancelation requests for persisted orders or quotes in that product will be rejected with reason "O: Unknown Order" until after the system restart completes. After the system restart, persisted orders/quotes can be cancelled from that time until the scheduled end of trading. In other words, the period of time in which persisted orders or quotes cannot be canceled starts at the scheduled end of trading for the associated product and ends after the system restarts. System restarts occur during a suspended state prior to the start of a queuing period and there may be minimal variation in the system restart time.

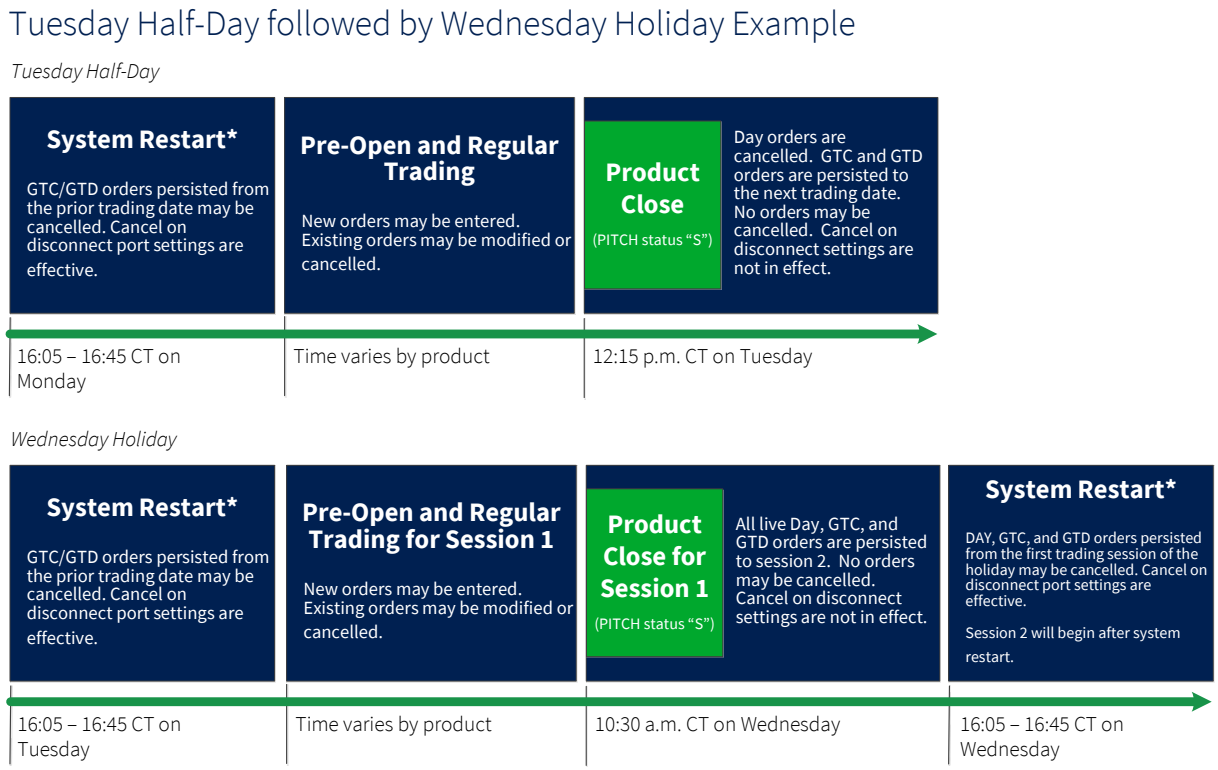
Regular Trading Example



Monday Holiday Example



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*The disconnect/reconnect sequence of a system restart generally takes about two minutes and could occur anytime between 16:05 and 16:45 CT.

1.5.4 Post-Settlement Execution Restatements

`Order Executed` messages received at the time of the trade in products VXT (Trade-At-Settlement ("TAS") for VX), VA (Variance Futures) and VAO (Variance Stub Futures) should be considered initial notification of trade. In all three of these products, information available only after the settlement time of the associated contract is required before the trade can be cleared. The following describe the post-settlement processing required for each applicable product:

VXT Execution prices of VXT (TAS) trades represent an offset to the end-of-day settlement price of the associated VX contract. For example, a trade executed at 0.02 is an agreement to buy and sell VX contracts at a price that 2-cents above the end-of-day settlement price, which is available after 3:00 p.m. CT. When VX end-of-day settlements are available, TAS trades executed during the business date are 'resolved' by updating the execution price and changing the symbol to the associated VX contract (TAS trades are cleared as VX trades).

VA Variance Futures are traded at prices in Volatility points (e.g., 15.5% volatility equals a price of 15.5) and quantity expressed in units of Vega (e.g., 100 equals 100 * 1,000 = 100,000 Vega, which is an exposure such that the value change of the position corresponding to a 1% change in Volatility change is $100,000). While VA trades are executed in Volatility and Vega terms, they are cleared in Variance price and size units. At the time of a trade, required information is available to compute the trade size in Variance units (i.e., traded size in Vega units, traded price in Volatility units, and expected and elapsed trading days). As a result, the pending `Order Execution` message at the time of execution, as well as the end-of-day `Variance Restatement` will contain the traded size in Variance units in the *ClearingSize* field. At approximately 4:00 p.m. CT, the closing price of the S&P 500 index is obtained and used to translate trade price and size to Variance units, after which trades can be cleared and restated.

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VAO Variance Stub futures are used to trade small size Variance Futures as required to exactly exit a previously entered Variance Futures position (see the Variance Futures Contract Specification). Exiting a VA position requires determining the number of Vega units to be transacted in order to offset a previously executed trade (note the Vega associated with a specified number of Variance Units changes daily). Inevitably, the associated Vega is not an even multiple of 1,000 (minimum VA contract size). To exit a VA position, the round lots of Vega are executed directly in VA. VAO trades are used to execute residual 'odd lots' of VA directly in Variance Units to completely exit a position. The 'odd lots' of Variance Units is computed by subtracting the Variance Units associated with the just executed offsetting VA trade (in Vega units) from the original size in Variance Units. Like VA futures, VAO trades in price units of Volatility. Unlike VA futures VAO futures trade directly in Variance units for size. As a result, both the pending `Order Execution` and the end-of-day `Variance Restatement` messages contain the *ClearingSize* populated with Variance units size, which is simply a copy of the *LastShares* field. At approximately 4:00 p.m. CT, the closing price of the S&P 500 index is obtained and used to translate trade price Variance units, after which trades can be cleared and restated as VA trades in the associated VA contract.

In all three of the above products, trades executed intraday are acknowledged back to participants using `Order Executed` messages. The `Order Executed` message received in these products is considered a 'Pending' trade. As a convenience to customers, an optional value PendingStatus is provided on the `Order Executed` message (see Section 7 - List of Optional Fields). CFE follows up each initial (i.e., pending) TAS and Variance future execution with post-settlement `TAS Restatement` and `Variance Restatement` messages respectively. The following summarizes the restatement details for each product:

VXT VXT trades are restated with the same *ExecID* and *ClOrdID* as the original trade. The as-executed symbol, price and size are maintained in the *Symbol*, *LastPx* and *LastShares* fields of the `TAS Restatement` message respectively. The VX symbol into which the TAS execution will clear (i.e., VX symbol with the same expiration as the as-executed VXT symbol) is contained in the *ClearingSymbol* field. The price with which the TAS execution will clear (i.e., the execution price offset with the contract settlement price) is contained in the *ClearingPrice* field.

VA VA trades are restated with the same *ExecID* and *ClOrdID* as the original trade. The as-executed symbol, price (in Volatility units) and size (in Vega units) are maintained in the *Symbol*, *LastPx* and *LastShares* fields of the `Variance Restatement` message respectively. The *NewSymbol* field will contain a copy of the as-executed Symbol since there is no symbol change for VA executions. The prices with which the VA execution will clear (i.e., the as-executed Volatility unit price in the *LastPx* field transformed to Variance units) is contained in the *ClearingPrice* field. Lastly, the size with which the VA execution will clear (i.e., the as-executed Vega unit price in the *LastShares* field transformed to Variance units) is contained in the *ClearingSize* field.

VAO VAO trades restated with the same *ExecID* and *ClOrdID* as the original trade. The as-executed symbol, price (in Volatility units) and size (in Variance units directly) are maintained in the *Symbol*, *LastPx* and *LastShares* fields of the `Variance Restatement` message respectively. The symbol into which the VAO execution will clear (i.e., the VA symbol with the same expiration as the as-executed VAO symbol) is contained in the *ClearingSymbol* field. The price with which the VAO execution will clear (i.e., the as-executed Volatility unit price transformed into Variance units) is contained in the *ClearingPrice* field. Lastly, the size with which the VAO execution will clear, which is the same as the *LastShares* field as the VAO instrument trades directly in Variance units, is contained in the *ClearingSize* field..

See sections 4.2.15 and 4.2.16 for details on the `TAS Restatement` and `Variance Restatement` messages used to restate TAS and Variance trades respectively.

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1.5.5 Spread Instruments and Signed Prices

All price fields in the CFE BOE protocol are signed values to accommodate spread instruments and TAS prices that can be negative (See 'Section 1.3 - Data Types' for a description and an example of using the *Binary Price* type, which is Little Endian byte order value, signed two's complement, eight bytes in size, with four implied decimal places). For an example of the use of the Binary Price type for negative price values, see the example BOE message in Section 4.2.15 - TAS Restatement. This section presents negative price scenarios introduced by Spread instruments.

Spreads instruments trade on CFE in a well-defined universe of two, three and four legged spreads with a restricted set of ratios and buy/sell conventions as shown in the table below. The notation S(1):B(1) means sell the first (earliest) expiration and buy the second (latest) expiration. The parenthesized numbers are the leg rations. For S(1):B(1) the ratios of each leg are 1, which means one unit of the spread contract is equivalent to selling 1 unit of the first expiration and buying 1 unit of the second expiration.

Legs	Spreads (B=Buy, S=Sell, ()=Ratio)
2	**S(1):B(1),** B(1):B(1), S(1):B(2), S(2):B(1)
3	B(1):B(1):B(1), B(1):S(2):B(1)
4	B(1):B(1):B(1):B(1), B(1):S(1):B(1):S(1), B(1):S(1):S(1):B(1)

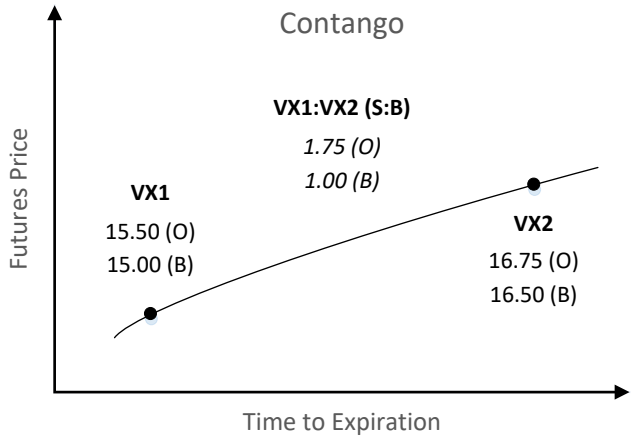
The bold 2-leg spread in the above table – S(1):B(1) – is a special spread that always exists in the CFE system. As new contracts are listed, the S(1):B(1) two leg spread instruments are automatically created between the new contract and all existing active contracts.

Spread instruments can result in executions where the buyer gets paid and the seller pays. This can be non-intuitive in all but the simplest spreads. Consider the two leg S(1):B(1) spread VX1:VX2 comprising selling 1 unit of the VX1 contract and buying 1 unit of the VX2 contract. To illustrate how buyers can get paid and sellers can pay, we examine spread pricing in Contango and Backwardation price environments.

Figure 1 below illustrates spread pricing in a 'Contango' price environment in which the price of the early expiration contract is lower than the later expiration contract. In this example the Bid/Offer of the VX1 simple contract is 15.00 x 15.50 and the Bid/Offer for the VX2 contract is 16.50 x 16.75. The synthetic market for the VX1:VX2 spread (i.e., the Bid/Offer implied by the leg markets) is 1.00 x 1.75. The bid of 1.00 derives from the fact that the offer on the VX1 leg is 15.50 and the bid on the VX2 leg is 16.50 and the net of the two is 1.00 net debit (i.e., buyer pays). Figure 1 shows the implied spread market in italics. This is the normal intuitive situation where the spread buyer pays and seller gets paid.

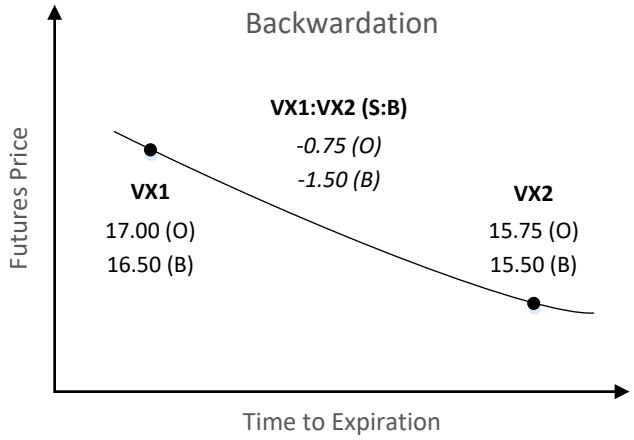
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Figure 1 - *Contango S(1):B(1) spread price example*



Next, consider the same example in the context of a Backward, or Inverted, market in which the price of the early expiration is higher than the price of the later expiration. Figure 2 below illustrates spread pricing in a Backward price environment. The Bid/Offer of the VX1 simple contract Is 16.50 x 17.00 and the Bid/Offer for the VX2 contract is 15.50 x 15.75. The synthetic market for the VX1:VX2 spread is -1.50 x -0.75. The bid of -1.50 derives from the fact that the offer on the VX1 leg is 17.00 and the bid on the VX2 leg is 15.50 and the net of the two is 1.50 net credit (i.e., buyer gets paid).

Figure 2 - *Backwardation (Inverted) S(1):B(1) spread price example*



Spread pricing requires thinking of instrument prices on the entire real number line and not just positive numbers. In the example above the bid is *less* than the offer as its left of the offer on the real number line. One can buy at the offer (paying -0.75 = receiving 0.75) and subsequently sell back at the bid (receiving -1.50 = paying 1.50), giving up the bid/offer spread (0.75) in the process; the same as positive prices. This concept generalizes to two and three leg spreads and unequal ratios; prices can just as easily be negative as positive as a result of the pricing environment (i.e., shape of the price curve vs. expiration date) and the spread definition (which legs bought/sold and ratios).

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1.5.6 OCC Clearing Reference

The following table can be used to assist firms in mapping values sent in BOE to their associated field names at the OCC. Note that *ClearingAccount* is not sent to the OCC.

BOE Field Name	FIX Tag	OCC Mapping
ClearingFirm	115	Exec Broker
Account	1	The first ten characters will appear in the Account # field. The entire 16 character string will appear in the Optional CM Data field.
ExecID	17	Trade ID
OrderId	37	Exchange Data
ClOrdId	11	Order ID
CMTANumber	439	CMTA CM#
ClearingAccount	440	Not sent to the OCC.

1.5.7 Port Types

All BOE port types may be ordered through the Customer Web Portal using the **Logical Port Request** tool. Changes to port attributes may be requested through the same tool by submitting a 'Modify' request for one or more existing BOE ports.

1.5.7.1 BOE Order Ports

Standard BOE ports (also referred to as order match capacity allocations) support simple and complex order entry as well as resetting TPH risk trips but do not support the usage of the following message types: `Quote Update`, `Purge Orders`. The attempted usage of any of these message types on standard BOE order ports will result in a rejection of the disallowed message.

Standard BOE ports are limited to 3,000 inbound messages per second. Once the inbound limit is reached new orders are rejected, modifies are handled as cancels, and cancels are processed normally. Standard BOE ports are limited to 10 inbound messages per second on CFE Test Products.

1.5.7.2 BOE Quoting Ports

BOE Quoting ports (also referred to as quoting match capacity allocations) are intended for use by firms quoting large numbers of simple futures contracts. BOE Quoting ports do not support the `Purge Orders` message type. This requires a separate BOE Purge Port (described below).

BOE Quoting ports provide efficiencies which may allow for reduced latency in comparison to the use of standard FIX and BOE ports. CFE will not operate more BOE Quoting ports than CPU cores on a server whereas this is not the case for BOE Order ports. In short, BOE Quoting ports are generally the more optimal port type to get an order/quote entered, modified, or cancelled in the matching engine as compared to standard FIX and BOE ports.

The *PreventMatch* field may not be specified on the `Quote Update` message and Match Trade Prevention is only available if defaulted at the port level. For Quoting ports, only Cancel Newest, Cancel Oldest, or Cancel Both are permitted. If a Bulk Quoting port is not configured with both a default MTP Modifier and Unique ID Level, Match Trade Prevention will be disabled.

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Quoting Port Order Acceptance Table

Message	Simple/Complex	Accepted over Bulk Quoting Port?	Other Conditions
`Quote Update`	Simple	Yes	Must be a TAS symbol
`Quote Update`	Complex	No	
`New Order`	Simple/Complex	Yes	Must have a *TimeInForce* value of Day, GTD, GTC, IOC, or FOK. Must be a TAS symbol
`Purge Orders`	Simple/Complex	No	
`Reset Risk`		Yes	

1.5.7.3 *BOE Purge Ports*

BOE Purge Ports support a single message type: `Purge Orders`. Members may use this port type to request a cancellation of groups of orders, including orders across multiple BOE/FIX Order or BOE Quoting ports. CFE will not operate more BOE or FIX Purge ports than CPU cores on a server.

1.5.8 Maximum Open Order Limits

The exchange limits the maximum number of open orders allowed on a BOE or BOE Quote port to 100,000 per port. New orders will be rejected once this limit is breached until the number of open orders drops back below 100,000. Note this limit is only for orders and does not include open quotes sent over a BOE Quote port.

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2 Session

2.1 Message Headers

Each message has a ten byte header. The two initial *StartOfMessage* bytes are present to aid in message reassembly for network capture purposes. The *MatchingUnit* field is only populated on sequenced, non-session level messages sent from CFE to the TPH. Messages from TPH to CFE and all session level messages must always set this value to 0.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	Message type.
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH. **For session level traffic, the unit is set to 0.** **For messages from TPH to CFE, the unit must be 0.**
SequenceNumber	6	4	Binary	The sequence number for this message. Messages from CFE to TPH are sequenced distinctly per matching unit. Messages from TPH to CFE are sequenced across all matching units with a single sequence stream. TPH can optionally send a 0 sequence number on all messages from TPH to CFE. CFE highly recommends that TPHs send sequence number on all inbound messages.

2.2 Login, Replay and Sequencing

Session level messages, both inbound (TPH to CFE) and outbound (CFE to TPH) are unsequenced.

Inbound (TPH to CFE) application messages are sequenced. Upon reconnection, CFE informs the TPH of the last processed sequence number; the TPH may choose to resend any messages with sequence numbers greater than this value. A gap forward in the TPH's incoming sequence number is permitted at any time and is ignored by CFE. Gaps backward in sequence number (including the same sequence number used twice) are never permitted and will always result in a `Logout` message being sent and the connection being dropped.

Most (but not all) outbound (CFE to TPH) application messages are monotonically sequenced per matching unit. Each message's documentation will indicate whether it is sequenced or unsequenced. While matching units on BOE correspond directly to matching units on Multicast PITCH, sequence numbers do not.

Upon reconnection, a TPH sends the last received sequence number per matching unit in a `Login Request` message. CFE will respond with any missed messages. However, when the `Login Request` *NoUnspeciedUnitReplay* flag is enabled, CFE will exclude messages from unspecified matching units during replay. CFE will send a `Replay Complete` message when replay is finished. If there are no messages to replay, a `Replay Complete` message will be sent immediately after a `Login Response` message. CFE will reject all orders during replay.

Assuming a TPH has requested replay messages using a properly formatted `Login Request` after a disconnect, any unacknowledged orders remaining with the TPH after the `Replay Complete` message is received should be assumed to be unknown to CFE.

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Unsequenced messages will not be included during replay.

A session is identified by the username and session sub-identifier (both supplied by CFE). Only one concurrent connection per username and session sub-identifier is permitted.

If a login is rejected, an appropriate `Login Response` message will be sent and the connection will be terminated.

2.3 Sequence Reset

A reset sequence operation is not available for Binary Order Entry. However, a TPH can send a `Login Request` message with *NoUnspecifiedUnitReplay* field enabled, and *NumberOfUnits* field set to zero. Then, upon receiving a `Login Response` message from CFE, the TPH can use the field *LastReceivedSequenceNumber* as the sequence starting point for sending future messages.

2.4 Heartbeats

`Client Heartbeat` messages are sent from TPH to CFE and `Server Heartbeat` messages are sent from CFE to TPH if no other data has been sent in that direction for one second. Like other session level messages, heartbeats from CFE to the TPH do not increment the sequence number. If CFE receives no inbound data or heartbeats for five seconds, a `Logout` message will be sent and the connection will be terminated. **TPHs are encouraged to have a one second heartbeat interval and to perform similar connection staleness logic.**

2.5 Logging Out

To gracefully log out of a session, a `Logout Request` message should be sent by the TPH. CFE will finish sending any queued data for that port and will then respond with its own `Logout` message and close the connection. After receipt of a `Logout Request` message, CFE will ignore all other inbound (TPH to CFE) messages except for `Client Heartbeat`.

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3 Session Messages

3.1 TPH to CFE

3.1.1 Login Request

A `Login Request` message must be sent as the first message upon connection.

A number of repeating parameter groups, some of which may be required, are sent at the end of the message. Ordering of parameter groups is not important. New parameter groups may be added in the future with no notice.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x37`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.
SessionSubID	10	4	Alphanumeric	Session Sub ID supplied by CFE.
Username	14	4	Alphanumeric	Username supplied by CFE.
Password	18	10	Alphanumeric	Password supplied by CFE.
NumberOfParam Groups	28	1	Binary	A number, n (possibly 0), of parameter groups to follow.
ParamGroup$_1$				First parameter group.
…				
ParamGroup$_n$				Last parameter group.

Unit Sequences Parameter Group

This parameter group includes the last consumed sequence number per matching unit received by the TPH. CFE uses these sequence numbers to determine what outbound (CFE to TPH) traffic, if any, was missed by the TPH. If this parameter group is not sent, it's assumed the TPH has not received any messages (e.g., start of day).

The TPH does not need to include a sequence number for a unit if they have never received messages from it. For example, if the TPH has received responses from units 1, 3, and 4, the `Login Request` message need not include unit 2. If the TPH wishes to send a value for unit 2 anyway, 0 would be the only allowed value.

Only one instance of this parameter group may be included.

Field	Offset	Length	Data Type	Description
ParamGroupLength	0	2	Binary	Number of bytes for the parameter group, including this field.
ParamGroupType	2	1	Binary	`0x80`
NoUnspecified UnitReplay	3	1	Binary	Flag indicating whether to replay missed outgoing (CFE to TPH) messages for unspecified units. `0x00` = False (Replay Unspecified Units) `0x01` = True (Suppress Unspecified Units Replay)

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NumberOfUnits	4	1	Binary	A number, n (possibly 0), of unit/sequence pairs to follow, one per unit from which the TPH has received messages.
UnitNumber 1		1	Binary	A unit number.
UnitSequence1		4	Binary	Last received sequence number for the unit.
…				
UnitNumber n		1	Binary	A unit number.
UnitSequencen		4	Binary	Last received sequence number for the unit.

Return Bitfields Parameter Group

This parameter group, which may be repeated, indicates which attributes of a message will be returned by CFE for the remainder of the session. This allows TPHs to tailor the echoed results to the needs of their system without paying for bandwidth or processing they do not need.

Listing of the return bitfields which are permitted per message is contained in 'Section 7 – Return Bitfields per Message'.

Field	Offset	Length	Data Type	Description
ParamGroupLength	0	2	Binary	Number of bytes for the parameter group, including this field.
ParamGroupType	2	1	Binary	`0x81`
MessageType	3	1	Binary	Return message type for which the bitfields are being specified (e.g., 0x25 for an `Order Acknowledgment` message).
NumberOfReturn Bitfields	4	1	Binary	Number of bitfields to follow.
ReturnBitfield1	5	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfieldn		1	Binary	Last bit field.

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Example Login Request Message:

Note this example is for illustrative purposes only. Actual login messages will contain specification of return bitfields for a larger set messages and each return bitfield specification will be complete whereas the example below is only an illustration for purposes of demonstrating the construction of the `Login Request` *message.*

Field Name	Hexadecimal	Notes
StartOfMessage	BA	Start of message bytes.
MessageLength	3D 00	61 bytes
MessageType	37	Login Request
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	00 00 00 00	Always 0 for session level messages
SessionSubID	30 30 30 31	0001
Username	54 45 53 54	TEST
Password	54 45 53 54 49 4E 47 00 00 00	TESTING
NumberOfParam Groups	03	3 parameter groups
ParamGroupLength	0F 00	15 bytes for this parameter group
ParamGroupType	80	0x80 = Unit Sequences
NoUnspecified UnitReplay	01	True (replay only specified units)
NumberOfUnits	02	Two unit/sequence pairs to follow;
UnitNumber₁	01	Unit 1
UnitSequence₁	4A BB 01 00	Last received sequence of 113,482
UnitNumber ₂	02	Unit 2
UnitSequence₂	00 00 00 00	Last received sequence of 0
ParamGroupLength	08 00	8 bytes for this parameter group
ParamGroupType	81	0x81 = Return Bitfields
MessageType	25	0x25 = Order Acknowledgment
NumberOfReturn Bitfields	03	3 bitfields to follow
ReturnBitfield₁	00	No bitfields from byte 1
ReturnBitfield₂	41	Symbol, Capacity
ReturnBitfield₃	05	Account, ClearingAccount
ParamGroupLength	0B 00	11 bytes for this parameter group
ParamGroupType	81	0x81 = Return Bitfields
MessageType	2C	0x2C = Order Execution
NumberOfReturn Bitfields	06	6 bitfields to follow
ReturnBitfield₁	00	No bitfields from byte 1
ReturnBitfield₂	41	*Symbol, Capacity*
ReturnBitfield₃	07	*Account, ClearingFirm, ClearingAccount*
ReturnBitfield₄	00	No bitfields from byte 4
ReturnBitfield₅	40	*BaseLiquidityIndicator*
ReturnBitfield₆	00	No bitfields from byte 6

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3.1.2 Logout Request

To end the session, the TPH should send a `Logout Request` message. CFE will finish sending any queued data and finally respond with a `Logout` message and close the connection.

A TPH may simply close the connection without logging out, but may lose any queued messages by doing so.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x02`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Logout Request Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`08 00`	8 bytes
MessageType	`02`	`Logout Request`
MatchingUnit	`00`	Always 0 for inbound messages
SequenceNumber	`00 00 00 00`	Always 0 for session level messages

3.1.3 Client Heartbeat

See 'Section 2.4 – Heartbeats' for more information about heartbeats and the session level protocol.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x03`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Client Heartbeat Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`08 00`	8 bytes
MessageType	`03`	`Client Heartbeat`
MatchingUnit	`00`	Always 0 for inbound messages
SequenceNumber	`00 00 00 00`	Always 0 for session level messages

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3.2 CFE to TPH

3.2.1 Login Response

`A Login Response` message is sent in response to a `Login Request` message. On a successful login, the *LoginResponseStatus* will be set to 'A'. On a failed login, *LoginResponseStatus* will be set to a value other than 'A', and *LoginResponseText* will be set to an appropriate failure description.

CFE will verify Return Bitfields at login time. If the Return Bitfields in a Return Bitfields Parameter Group are invalid, *LoginResponseStatus* will be set to F, and *LoginResponseText* will include a description of which byte and bit are invalid. This is done to ensure that reserved fields are not used, and only options that apply to the local market are set. See 'Section 6 - Return Bitfields Per Message' for additional information.

Note that two sets of sequence numbers are available on the `Login Response`. The set of sequence numbers in the body are the actual CFE to TPH sequence numbers indicating the highest sequence numbers available per matching unit. If specified during login, the Unit Sequences Parameter Group will also be returned which is an echo of the sequence numbers the TPH presented during login as the highest received. If these are different, it indicates a gap which will be filled by CFE.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x24`
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.
LoginResponseStatus	10	1	Alphanumeric	Accepted, or the reason for the rejection. `A` = Login Accepted `N` = Not authorized (invalid username/pwd) `D` = Session is disabled `B` = Session in use `S` = Invalid session `Q` = Sequence ahead in Login message `I` = Invalid unit given in Login message `F` = Invalid return bit field in login message `M` = Invalid Login Request message structure
LoginResponseText	11	60	Text	Human-readable text with additional information about the reason for rejection. ASCII NUL (0x00) filled on the right, if necessary.
NoUnspecified UnitReplay	71	1	Binary	Echoed back from the original `Login Request` message.
LastReceived SequenceNumber	72	4	Binary	Last inbound (TPH to CFE) message sequence number processed by CFE.
NumberOfUnits	76	1	Binary	A number, n, of unit/sequence pairs to follow, one per unit. A pair for every unit will be sent, even if no messages have been sent to this port today. For unsuccessful logins, this will be 0.
UnitNumber $_1$		1	Binary	A unit number.
UnitSequence$_1$		4	Binary	Highest available CFE to TPH sequence number for the unit.
…				
UnitNumber $_n$		1	Binary	A unit number.

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UnitSequence$_n$		4	Binary	Highest available CFE to TPH sequence number for the unit.
NumberOfParam Groups		1	Binary	Echoed back from the original `Login Request` message.
ParamGroup$_1$				Echoed back from the original `Login Request message`.
…				
ParamGroup$_n$				Echoed back from the original `Login Request` message.

Example Login Response Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`78 00`	120 bytes
MessageType	`24`	`Login Response`
MatchingUnit	`00`	Always 0 for session messages
SequenceNumber	`00 00 00 00`	Always 0 for session level messages
LoginResponseStatus	`41`	`A` = Login Accepted
LoginResponseText	`41 63 63 65 70 74 65 64 00 00`	`Accepted`
	`00 00 00 00 00 00 00 00 00 00`	(padding)
	`00 00 00 00 00 00 00 00 00 00`	(padding)
	`00 00 00 00 00 00 00 00 00 00`	(padding)
	`00 00 00 00 00 00 00 00 00 00`	(padding)
	`00 00 00 00 00 00 00 00 00 00`	(padding)
NoUnspecified UnitReplay	`01`	True (replay only specified units)
Last Received Sequence Number	`54 4A 02 00`	Last sequence CFE received of 150,100
NumberOfUnits	`02`	Two unit/sequence pairs to follow;
UnitNumber 1	`01`	Unit 1
UnitSequence1	`4A BB 01 00`	Actual last sequence of 113,482
UnitNumber 2	`02`	Unit 2
UnitSequence2	`00 00 00 00`	Actual last sequence of 0
NumberOfParam Groups	`03`	3 parameter groups
ParamGroupLength	`0E 00`	15 bytes for this parameter group
ParamGroupType	`80`	`0x80` = Unit Sequences
NoUnspecified UnitReplay	`01`	True (replay unspecified units)
NumberOfUnits	`02`	Two unit/sequence pairs to follow;
UnitNumber 1	`01`	Unit 1
UnitSequence1	`4A BB 01 00`	Last received sequence of 113,482
UnitNumber 2	`02`	Unit 2
UnitSequence2	`00 00 00 00`	Last received sequence of 0
ParamGroupLength	`08 00`	8 bytes for this parameter group
ParamGroupType	`81`	0x81 = Return Bitfields
MessageType	`25`	0x25 = `Order Acknowledgment`

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NumberOfReturn Bitfields	03	3 bitfields to follow
ReturnBitfield1	00	No bitfields from byte 1
ReturnBitfield2	41	Symbol, Capacity
ReturnBitfield3	05	Account, ClearingAccount
ParamGroupLength	0B 00	11 bytes for this parameter group
ParamGroupType	81	0x81 = Return Bitfields
MessageType	2C	0x2C = Order Execution
NumberOfReturn Bitfields	06	6 bitfields to follow
ReturnBitfield1	00	No bitfields from byte 1
ReturnBitfield2	41	*Symbol, Capacity*
ReturnBitfield3	07	*Account, ClearingFirm, ClearingAccount*
ReturnBitfield4	00	No bitfields from byte 4
ReturnBitfield5	40	*BaseLiquidityIndicator*
ReturnBitfield6	00	No bitfields from byte 6

3.2.2 Logout

A Logout is usually sent in response to a Logout Request. Any queued data is transmitted, a Logout is sent, and CFE will close the connection. However, a Logout may also be sent if the TPH violates the protocol specification (e.g., by moving backwards in sequence number).

The Logout contains the last transmitted sequence number for each unit, allowing the TPH to check that their last received sequence number matches.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x08
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.
LogoutReason	10	1	Alphanumeric	The reason why the Logout message was sent. U = User Requested E = End of Day A = Administrative ! = Protocol Violation
LogoutReasonText	11	60	Text	Human-readable text with additional information about the reason for logout. Particularly useful if *LogoutReason* = ! (Protocol Violation).
LastReceived SequenceNumber	71	4	Binary	Last inbound (TPH to CFE) message sequence number processed by CFE.
NumberOfUnits	75	1	Binary	A number, *n* (possibly 0), of unit/sequence pairs to follow, one per unit from which the client has received messages.
UnitNumber 1		1	Binary	A unit number.
UnitSequence1		4	Binary	Highest available sequence number for the unit.
…				
UnitNumber n		1	Binary	A unit number.

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UnitSequence$_n$		4	Binary	Highest available sequence number for the unit.

Example Logout Response Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	54 00	84 bytes
MessageType	08	Logout
MatchingUnit	00	Always 0 for session level messages
SequenceNumber	00 00 00 00	Always 0 for session level messages
LogoutReason	55	U = User Requested
LogoutReasonText	55 73 65 72 00 00 00 00 00 00	User
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
	00 00 00 00 00 00 00 00 00 00	
LastReceived SequenceNumber	54 5A 02 00	Last CFE received sequence of 150,100
NumberOfUnits	02	Two unit/sequence pairs to follow;
UnitNumber $_1$	01	Unit 1
UnitSequence$_1$	4A BB 01 00	Last sent sequence of 113,482
UnitNumber $_2$	02	Unit 2
UnitSequence$_2$	00 00 00 00	Last sent sequence of 0

3.2.3 Server Heartbeat

See 'Section 2.4 - Heartbeats' for more information about heartbeats and the session level protocol.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x09
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Server Heartbeat Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	08 00	8 bytes
MessageType	09	Server Heartbeat
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	00 00 00 00	Always 0 for session level messages

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3.2.4 Replay Complete

See 'Section 2.2 - Login, Replay and Sequencing' for more information on Login, sequencing and replay.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x13`
MatchingUnit	5	1	Binary	Always 0 for session level messages.
SequenceNumber	6	4	Binary	Always 0 for session level messages.

Example Replay Complete Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`08 00`	8 bytes
MessageType	`13`	`Replay Complete`
MatchingUnit	`00`	Always 0 for inbound messages
SequenceNumber	`00 00 00 00`	Always 0 for session level messages

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4 Application Messages

4.1 TPH to CFE

4.1.1 New Order

A `New Order` message consists of a number of required fields followed by a number of optional fields. The optional fields used are specified by setting bits in the *NewOrderBitfields*. Fields must be appended at the end of the message, starting with the lowest order enabled bit in the first bit field first.

Permitted input optional fields are described in 'Section 5.1 – New Order'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x38`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
ClOrdID	10	20	Text	Corresponds to ClOrdID (11) in CFE FIX. Unique ID chosen by the client. Characters in the ASCII range 33-126 are allowed, except for comma, semicolon, and pipe. A leading tilde (~) cannot be sent on any ClOrdId and will result in a reject. These are reserved for internal use by CFE and could be received as a result of a system-generated ClOrdId. If the ClOrdID matches a live order, the order will be rejected as duplicate. Sent to the OCC in the Order ID field. Note: CFE only enforces uniqueness of ClOrdID values among currently live orders, which includes long-lived GTC and GTD orders. However, we strongly recommend that you keep your ClOrdID values unique.
Side	30	1	Alphanumeric	Corresponds to *Side* (54) in CFE FIX. 1 = Buy 2 = Sell
OrderQty	31	4	Binary	Corresponds to *OrderQty* (38) in CFE FIX. Order quantity. System limit is 999,999 contracts.
NumberOf NewOrder Bitfields	35	1	Binary	Bitfield identifying which bitfields are set. Field values must be appended to the end of the message.
NewOrderBitfield¹	36	1	Binary	Bitfield identifying fields to follow.
….				
NewOrderBitfieldⁿ		*1*	*Binary*	*Last bitfield.*
Optional fields. . .				

Example New Order Message:

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Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	64 00	100 bytes
MessageType	38	New Order
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	64 00 00 00	Sequence number 100
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
Side	31	1 = Buy
OrderQty	64 00 00 00	100 contracts
NumberOfNewOrder Bitfields	09	9 bitfields to follow
NewOrderBitfield1	34	*Price, OrdType, TimeInForce*
NewOrderBitfield2	41	*Symbol, Capacity*
NewOrderBitfield3	01	*Account*
NewOrderBitfield4	10	*OpenClose*
NewOrderBitfield5	00	*(none)*
NewOrderBitfield6	00	*(none)*
NewOrderBitfield7	E0	*CtiCode, ManualOrderIndicator, OEOID*
NewOrderBitfield8	00	*(none)*
NewOrderBitfield9	10	*CustOrderHandlingInst*
Price	F0 49 02 00 00 00 00 00	15.00
OrdType	32	2 = Limit
TimeInForce	30	0 = Day
Symbol	30 30 30 30 30 37 00 00	000007
Capacity	43	C = Customer
Account	30 30 32 00 00 00 00 00 00 00 00 00 00 00 00 00	002
OpenClose	4F	O = Open
CtiCode	31	1 = TPH for TPH Acct
ManualOrderIndicator	59	Y = Manual
OEOID	4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00	JOHN DOE
CustOrderHandlingInst	57	W = Desk

4.1.2 Cancel Order

Request to cancel either a single order or mass cancellation of a group of orders or quotes.

- A single order cancellation uses the *ClOrdID* from a previous order (*OrigClOrdID* field)
- Mass cancellation of a group of orders or quotes requires sending *MassCancelInst* which comprises filters used to specify the set of orders to cancel.
    - If the Clearing Firm Filter is set to "F", the *ClearingFirm* optional field must be specified or the Cancel Order request will be rejected.
    - If the Acknowlegement Style is set to "S" or "B", or in combination with "M", the *MassCancelID* optional field must be specified or the `Cancel Order` request will be rejected.
    - If the *ProductName* optional field only orders for instruments associated with the product (e.g., "VX") are cancelled.
    - If Lockout Instruction is set to "L" and the *ProductName* optional field is not specified, a *Firm-level* Risk Reset is required to clear the Lockout condition. If Lockout Instruction is set to "L" and the *ProductName* optional field is specified, a Product level reset is required. See specification of the RiskReset optional field in **List of Optional Fields**.

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- Lockout will apply to all `New Order` and `Modify Order` messages for the *ClearingFirm* (and *ProductName* if specified), regardless of other filtering in the cancel order request message.

The *ManualOrderIndicator and OEOID* fields in the optional field block must be present on all `Cancel Order` requests. Messages sent without these fields will be rejected.

The system limits the rate at which identical mass cancel and `Purge Orders` requests can be submitted to the system. Requests are restricted to twenty (20) messages per second per port.

An identical mass cancel message is defined as a message having all of the same *CustomGroupID*, *Symbol*, *Clearing Firm*, *Lockout Instruction*, *Instrument Type Filter* and *GTC Order Filter* field values, as a previously received message.

Permitted input optional fields are described in 'Section 5.2 – Cancel Order'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x39`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
OrigClOrdID	10	20	Text	Corresponds to *OrigClOrdID* (41) in CFE FIX. *ClOrdID* of the order to cancel. For mass cancel requests, must be empty (all zeroes).
NumberOf CancelOrder Bitfields	30	1	Binary	Bitfield identifying bitfields which are set. May be 0. Field values must be appended to the end of the message.
CancelOrder Bitfield[1]	31	1	Binary	Bitfield identifying fields to follow. Only present if *NumberOfCancelOrderBitfields* is non-zero.
…				
CancelOrder Bitfield[n]		1	Binary	Last bitfield.
Optional fields. . .				

Example Cancel Order Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`35 00`	53 bytes
MessageType	`39`	`Cancel Order`
MatchingUnit	`0`	Always 0 for inbound messages
SequenceNumber	`64 00 00 00`	Sequence Number 100
OrigClOrdID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00`	ABC123
NumberOfCancel OrderBitfields	`01`	1 bitfield to follow
CancelOrderBitfield1	`C1`	*ClearingFirm, ManualOrderIndicator, OEOID*
ClearingFirm	`54 45 53 54`	TEST
ManualOrderIndicaator	`59`	Y = Manual
OEOID	`4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00 00 00`	JOHN DOE

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Example Mass Cancel Order Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	60 00	96 bytes
MessageType	39	Cancel Order
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	64 00 00 00	Sequence Number 100
OrigClOrdID	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	(empty)
NumberOfCancel OrderBitfields	02	2 bitfields to follow
CancelOrderBitfield1	D9	*ClearingFirm, ProductName, MassCancelID, ManualOrderIndicator, OEOID*
CancelOrderBitfield2	01	*MassCancelInst*
ClearingFirm	54 45 53 54	TEST
ProductName	56 58 00 00 00 00	VX
MassCancelID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
ManualOrderIndicaator	59	Y = Manual
OEOID	4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00	JOHN DOE
MassCancelInst	46 4D 4E 42 50 00 00 00 00 00 00 00 00 00 00 00	FMNBP

4.1.3 Modify Order

Request to modify an order. The order attributes to be modified are selected using *NumberOfModifyBitfields* and some number of bitfields to follow. *Price*, *OrderQty*, *OrdType*, *StopPx*, *ManualOrderIndicator*, *CustOrderHandlingInst*, and *OEOID* may be adjusted. Modifies will result in a loss of time priority unless (1) they have no change in *Price* and also reduce *OrderQty* or (2) they change the *StopPx* for a stop order that has not been elected. *OrdType* may be adjusted from Limit to Market.

Changes in *OrderQty* result in an adjustment of the current order's *OrderQty*. The new *OrderQty* does not directly replace the current order's *LeavesQty*. Rather, a delta is computed from the current *OrderQty* and the replacement *OrderQty*. This delta is then applied to the current *LeavesQty*. If the resulting *LeavesQty* is less than or equal to zero, the order is cancelled. This results in safer behavior when the modification request overlaps partial fills for the current order, leaving the TPH in total control of the share exposure of the order.

A `Modify Order` should not be issued until the `Order Acknowledgement` for the previous `New Order` or `Order Modified` message for the previous `Modify Order` has been received. The BOE handler will reject a new `Modify Order` if it has not been accepted or it has not seen the result of the prior modification from the Matching Engine. However, `Modify Order` requests that merely reduce *OrderQty* may be overlapped if the existing *ClOrdID* is reused. This is the only case where reuse of the *ClOrdID* is allowed. A leading tilde (~) cannot be sent on any *ClOrdId* and will result in a reject. These are reserved for internal use by CFE and could be received as a result of a system-generated *ClOrdId*.

The *OrderQty, Price, ManualOrderIndicator and OEOID* fields in the optional field block must be present on all `Modify Order` requests. Messages sent without these fields will be rejected.

A maximum of 1,679,615 `Modify Order` requests may be made to a single order each trading day. Once the 1,679,615th modification is made, then the next user-generated message on the order should be a `Cancel Order` request.

Permitted input optional fields are described in 'Section 5.3 – Modify Order'.

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Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x3A`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
ClOrdID	10	20	Text	New *ClOrdID* for this order. Unique ID chosen by the client. Characters in the ASCII range 33-126 are allowed, except for comma, semicolon, and pipe. A leading tilde (~) cannot be sent on any ClOrdId and will result in a reject. These are reserved for internal use by CFE and could be received as a result of a system-generated ClOrdId. If the ClOrdID matches a live order, the order will be rejected as duplicate. Sent to the OCC in the Order ID field. Note: CFE only enforces uniqueness of ClOrdID values among currently live orders, which includes long-lived GTC and GTD orders. However, we strongly recommend that you keep your ClOrdID values unique.
OrigClOrdID	30	20	Text	Corresponds to *OrigClOrdID* (41) in CFE FIX. *ClOrdID* of the order to replace. In the case of multiple changes to a single order, this will be the *ClOrdID* of the most recently accepted change.
NumberOf ModifyOrder Bitfields	50	1	Binary	Bitfield identifying bitfields which are set. May be 0. Field values must be appended to the end of the message.
ModifyOrder Bitfield1	51	1	Binary	Bitfield identifying fields to follow.
…				
ModifyOrder Bitfieldn		1	Binary	Last bitfield.
Optional fields. . .				

Example Modify Order Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes
MessageLength	`53 00`	83 bytes
MessageType	`3A`	`Modify Order`
MatchingUnit	`00`	Always 0 for inbound messages
SequenceNumber	`64 00 00 00`	Sequence Number 100
ClOrdID	`41 42 43 31 32 34 00 00 00 00 00 00 00 00 00 00 00 00 00 00`	ABC124

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OrigClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
NumberOfModify OrderBitfields	02	2 bitfields to follow
ModifyOrderBitfield1	0C	*OrderQty, Price*
ModifyOrderBitfield2	58	*ManualOrderIndicator, OEOID, CustOrderHandlingInst*
OrderQty	64 00 00 00	100 contracts
Price	08 E2 01 00 00 00 00 00	12.34
ManualOrderIndicaator	59	Y = Manual
OEOID	4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00 00 00	JOHN DOE
CustOrderHandlingInst	59	Y = Electronic

4.1.4 Quote Update

Request to enter or update one or more quotes. `Quote Update` requests will be forwarded in their entirety to the matching engine instance as a single message and will be applied in a single transaction. Optional bitfields are not supported for any response messages for quotes.

All contracts in a single `Quote Update` must trade under a single futures root. Requests which include contracts trading under multiple futures roots will be rejected in their entirety.

A quote is unique per port, EFID, and side. You may quote multiple price levels of depth using either multiple EFIDs on a single port or with the same EFID on multiple ports.

Quote requests are one-sided. The cancellation of quotes can be done by sending a `Quote Update` with a zero size and a *SizeModifier* of NULL (0x00). The table below describes the possible combinations of zero/non-zero sizes and prices and the resulting behavior. The behavior differs depending on the product type (TAS vs non-TAS) and the value of the *SizeModifier* field.

Product Type	Zero Size	Zero Price	*SizeModifier*	Result
Trade At Settlement (“TAS”)	Y	Y	0	Quote is cancelled
	Y	N	0	Quote is cancelled
	N	Y	0	Quote price updated
	Y	Y	R	Quote price updated
	Y	N	R	Quote price updated
	N	Y	R	Quote size and price updated
Standard (non-TAS)	Y	Y	0	Quote is cancelled
	Y	N	0	Quote is cancelled
	N	Y	0	Quote is cancelled
	Y	Y	R	Quote is cancelled
	Y	N	R	Quote price updated
	N	Y	R	Quote is cancelled

Quotes may be for simple instruments only; complex/spread quotes may not be submitted.

All quotes will be automatically cancelled at the end of the trading day.

If a quote modification is rejected, the resting quote being modified is also cancelled.

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Executions, unsolicited cancels, and unsolicited modification response messages from the exchange are different from those for orders. They are optimized for efficiency and contain some different data elements (e.g., *QuoteUpdateID*) than the respective messages for orders.

The *PreventMatch* field may not be specified on the `Quote Update` message and Match Trade Prevention is only available if defaulted at the port level. For BOE Quoting ports, only Cancel Newest, Cancel Oldest, or Cancel Both are permitted. If a BOE Quoting port is not configured with both a default MTP Modifier and Unique ID Level, Match Trade Prevention will be disabled.

To maintain time priority, all attributes of an existing quote *must* be unchanged except for a reduction in size. Changing any other attribute or increasing size will result in a loss of time priority.

Capacity may not be changed when modifying a quote. To change *Capacity* of a resting quote, you must first send a quote with zero price and size and then re-enter the quote with the desired *Capacity*.

`Quote Update` and `New Order` messages submitted through BOE Quoting ports will be available over ODROP.

During the queuing period, a `Quote Update` may only contain a bid and/or offer for a single TAS expiration.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x70`
MatchingUnit	5	1	Binary	Always 0 for inbound (Member to CFE) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
QuoteUpdateID	10	16	Text	ID chosen by the client. Characters in the ASCII range 33-126 are allowed, except for comma, semicolon, and pipe. All quote response messages will include this identifier. Note: CFE only enforces uniqueness of *QuoteUpdateID* values among those not yet acknowledged by the ME. However, we strongly recommend that you keep your *QuoteUpdateID* values unique for a trading day.
ClearingFirm	26	4	Alpha	EFID that will clear the trade. If left blank, the port attribute value of 'Default Executing Firm ID' is used.
ClearingAccount	30	4	Alpha	Corresponds to *OnBehalfOfSubID* (116) and *ClearingAccount* (440) in CFE FIX. See **List of Optional Fields** for additional information.
CMTANumber	34	4	Binary	Corresponds to *CMTANumber* (439) in CFE FIX. See **List of Optional Fields** for additional information.
Account	38	16	Text	Corresponds to *Account* (1) in CFE FIX. See **List of Optional Fields** for additional information.
CustomGroupID	54	2	Binary	Optional. Used to group orders for use in `Purge Orders`. Set to 0 if functionality not needed.
Capacity	56	1	Alpha	Corresponds to *OrderCapacity* (47) in CFE FIX. See **List of Optional Fields** for additional information.
CtiCode	57	1	Alphanumeric	Corresponds to *CTICode* (9702) in CFE FIX.

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				See **List of Optional Fields** for additional information.
ManualOrder Indicator	58	1	Alpha	Corresponds to *ManualOrderIndicator* (1028) in CFE FIX. See **List of Optional Fields** for additional information.
OEOID	59	18	Text	Corresponds to OEOID (25004) in CFE FIX. See **List of Optional Fields** for additional information.
SizeModifier	77	1	Text	Controls the behavior of the *OrderQty* field. Using "R" allows for a TPH to ensure that in-flight fills or cancels do not result in unwanted additional size exposure. `NULL (0x00)` = New quote size will be set to value of *OrderQty*. R = Reduce outstanding size of quote by the *OrderQty* provided. When using "R", if the resulting size is zero or negative, then the quote is cancelled. TPHs are expected to track the remaining quantity of each quote as resulting size is not included on the `Quote Update Acknowledgement` message.
Reserved	78	6	Binary	Reserved for future expansion. To maintain forward compatibility, fill with 0.
QuoteCnt	84	1	Binary	Number of repeating groups included in this quote update. Allowed values are 1-20.
Repeating Groups of …				
Symbol		6	Alphanumeric	CFE native identifier
Side		1	Text	1 = Buy 2 = Sell
OpenClose		1	Text	Corresponds to *OpenClose* (77) in CFE FIX. See **List of Optional Fields** for additional information.
Price		8	Binary Price	Limit price. To cancel an existing quote, specify a size of 0.
OrderQty		4	Binary	Order quantity or quantity to reduce if *SizeModifier* = R. System limit is 999,999 contracts.
Reserved		12	Binary	Reserved for future expansion. To maintain forward compatibility, fill with 0.

Example Quote Update Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`93 00`	147 bytes
MessageType	`70`	`Quote Update`
MatchingUnit	`00`	Always 0 for inbound messages
SequenceNumber	`64 00 00 00`	Sequence number 100
QuoteUpdateID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00`	ABC123
ClearingFirm	`41 42 43 44`	ABCD

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ClearingAccount	57 58 59 5A	WXYZ
CMTANumber	31 32 33 34	1234
Account	44 45 46 47 41 42 43 44 00 00 00 00 00 00 00 00	DEFGABCD
CustomGroupID	C8 00	200
Capacity	46	F = Firm
CTICode	32	2 = TPH
ManualOrderIndicator	59	Y = Manual
OEOID	4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00	JOHN DOE
Reserved	00 00 00 00 00 00 00	Reserved
QuoteCnt	02	Two Quotes
Symbol	30 30 36 69 70 41	006ipA
Side	31	1 = Buy
OpenClose	4F	O = Open
Price	C8 32 00 00 00 00 00 00	1.30
OrderQty	64 00 00 00	100 contracts
SizeModifier	00	NULL = use *OrderQty* provided as quote size
Reserved	00 00 00 00 00 00 00 00 00 00 00	Reserved
Symbol	30 30 34 63 53 73	004cSs
Side	32	2 = Sell
OpenClose	4F	O = Open
Price	AC 07 01 00 00 00 00 00	6.75
OrderQty	F4 01 00 00	500 contracts
Reserved	00 00 00 00 00 00 00 00 00 00 00 00	Reserved

4.1.5 Purge Orders

Request to cancel a group of orders or quotes across all the TPH's sessions. This differs from a mass cancel request sent via a `Cancel Order` message as the purge request is applied across all of the TPH's sessions, not just the session on which the `Cancel Order` was received. In addition, the `Purge Orders` message accepts a list of *CustomGroupIDs* as part of the order matching filter.

- `Purge Orders` requires sending *MassCancelInst* bitfield.
- Optionally *ProductName, ClearingFirm, MassCancelID* and list of *CustomGroupID* may also be sent.
- *ProductName* and *CustomGroupID* are mutually exclusive. Messages containing both will be rejected.
- A maximum of 10 *CustomGroupID* may be sent in one message.
- A `Mass Cancel Acknowledgment` message may be requested by setting the Acknowledgement Style value in the required 'optional' field *MassCancelInst* to "S" or "B"or in combination with "M". In these cases, the Purge Orders request will be rejected if the *MassCancelID* optional field is not provided.
- Individual `Order Cancelled` or `Quote Cancelled` messages are requested by setting the Acknowledgement Style value of the required 'optional' field *MassCancelInst* to "M" or "B".

The *ManualOrderIndicator and OEOID* fields in the optional field block must be present on all `Purge Orders` requests. Messages sent without these fields will be rejected.

The system limits the rate at which identical Mass Cancel and Purge Orders requests can be submitted to the system. Requests are restricted to twenty (20) messages per second per port.

An identical Mass Cancel message is defined as a message having all of the same *CustomGroupID*, *Symbol*, *Clearing Firm*, *Lockout Instruction*, *Instrument Type Filter* and *GTC Order Filter* field values, as a previously received message.

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Permitted input optional fields are described in 'Section 5.4 – Purge Orders'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x47`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
ReservedInternal	10	1	Alphanumeric	Reserved for CFE Internal use
NumberOf PurgeOrders Bitfields	11	1	Binary	Bitfield identifying bitfields which are set. May be 0. Field values must be appended to the end of the message.
PurgeOrders Bitfield1	12	1	Binary	Bitfield identifying fields to follow. Only present if *NumberOfPurgeOrdersBitfields* is non-zero.
CustomGroupIDCnt	13	1	Binary	Number of repeating *CustomGroupID* included in this message.
CustomGroupIDOne		2	Binary	First *CustomGroupID*. Only present if *CustomGroupIDCnt* is non-zero.
…				
CustomGroupIDN		2	Binary	Last *CustomGroupID*.
Optional fields. . .				

Example Purge Orders Message with CustomGroupID and Firm Level Lockout:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes
MessageLength	`4B 00`	75 bytes
MessageType	`47`	`Purge Orders`
MatchingUnit	`0`	Always 0 for inbound messages
SequenceNumber	`64 00 00 00`	Sequence Number 100
Reserved	`00`	Ignore
NumberOfPurge OrderBitfields	`01`	1 bitfield to follow
PurgeOrdersBitfield1	`D5`	*ClearingFirm, MassCancelInst, MassCancelID, ManualOrderIndicator, OEOID*
CustomGroupIDCnt	`02`	2 CustomGroupIDs to follow
CustomGroupID1	`BF BE`	First *CustomGroupID* of 48831
CustomGroupID2	`CO BE`	Second *CustomGroupID* of 48832
ClearingFirm	`54 45 53 54`	`TEST`
MassCancelInst	`46 42 4C 42 44 00 00 00 00 00 00 00 00 00 00 00`	FBLBD
MassCancelID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00`	ABC123
ManualOrderIndicator	`59`	Y = Manual
OEOID	`4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00`	JOHN DOE

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Example Purge Orders Message with Product Level Filter and no Lockout:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes
MessageLength	4D 00	77 bytes
MessageType	47	Purge Orders
MatchingUnit	00	Always 0 for inbound messages
SequenceNumber	64 00 00 00	Sequence Number 100
Reserved	00	Ignore
NumberOfPurge OrderBitfields	01	1 bitfield to follow
PurgeOrdersBitfield1	DD	*ClearingFirm, MassCancelInst, ProductName, MassCancelID, ManualOrderIndicator, OEOID*
CustomGroupIDCnt	00	No *CustomGroupID* to follow
ClearingFirm	54 45 53 54	TEST
MassCancelInst	46 42 4E 42 43 00 00 00 00 00 00 00 00 00 00 00	FBNBC
ProductName	56 58 00 00 00 00	VX
MassCancelID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
ManualOrderIndicaator	59	Y = Manual
OEOID	4A 4F 48 4E 20 44 4F 45 00 00 00 00 00 00 00 00 00 00	JOHN DOE

4.1.6 Reset Risk

Reset or release Firm/EFID, Product, or Custom Group ID level lockout conditions resulting from risk profile trips or self-imposed lockouts issued via `Cancel Order` or `Purge Orders` messages. Risk resets can be performed using this message or by using the *RiskReset* field on a `New Order` message.

Only one unique risk reset of a given type (Firm/EFID, Product, CustomGroupID) is allowed per second. Additional resets will be ignored (*RiskResetResult* = <space>). For example, a customer may reset risk for *CustomGroupID* = 1 and may not reset risk again for *CustomGroupID* = 1 until one second has elapsed. This restriction is designed to safeguard the trading platform from excessive risk messaging.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x56`
MatchingUnit	5	1	Binary	Always 0 for inbound (TPH to CFE) messages.
SequenceNumber	6	4	Binary	The sequence number for this message.
RiskStatusID	10	16	Text	Unique identifier for this Reset Risk request. Response message will have this corresponding identifier. **Note: CFE only enforces uniqueness of *RiskStatusID* values among currently unacknowledged requests. However, we strongly recommend that you keep your *RiskStatusID* values day-unique.**

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RiskReset	26	8	Text	Corresponds to *RiskReset* (7692) in CFE FIX. Indicates Firm, Product, or CustomGroupID lockout reset. See **List of Optional Fields** for allowed values.
Reserved	34	4	Binary	Reserved for future expansion. To maintain forward compatibility, fill with 0.
ClearingFirm	38	4	Alpha	Risk will be reset for this EFID. Resets a self-imposed EFID-level lockout initiated using a mass cancel or purge request. Required on all resets.
ProductName	42	6	Text	Populate with Futures product name for resets at the Futures Root/product level. For example, use to reset and allow new order/quotes if a TPH trips a traded volume rate limit for futures product or has initiated a self-imposed lockout at the product level. Leave empty for resets at the EFID level only.
CustomGroupID	48	2	Binary	Populate with a valid *CustomGroupID* to reset a self-imposed *CustomGroupID* lockout initiated with a `Purge Orders` request. Set to 0 to ignore or for resets at the EFID level only.

Example Reset Risk Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`30 00`	48 bytes
MessageType	`56`	`Reset Risk`
MatchingUnit	`00`	Always 0 for inbound messages
SequenceNumber	`64 00 00 00`	Sequence number 100
RiskStatusID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00`	ABC123
RiskReset	`53 46 00 00 00 00 00 00`	SF = Product and Firm level reset
Reserved	`00 00 00 00`	
ClearingFirm	`54 45 53 54`	TEST
RiskRoot	`56 58 00 00 00 00`	VX
CustomGroupID	`00 00`	No *CustomGroupID*

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4.2 CFE to TPH

4.2.1 Order Acknowledgment

`Order Acknowledgment` messages are sent in response to a `New Order` message. The message corresponds to a FIX Execution Report with *ExecType* (150) = 0 (New).

Per the instructions given in a Return Bitfields Parameter Group on the `Login Request` (Section 3.1.1 – Login Request), optional fields may be appended to echo back information provided in the original `New Order` message. Fields which have been requested to be echoed back but which were not filled in will still be sent, but filled with binary zero (`0x00`).

Permitted return optional fields are described in 'Section 6.1 – Order Acknowledgement'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	`0x25`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	18	20	Text	Echoed back from the original order.
OrderID	38	8	Binary	Corresponds to *OrderID* (37) in CFE FIX. Order identifier supplied by CFE. This identifier corresponds to the identifiers used in CFE market data products. Sent to the OCC in the Exchange Data field.
ReservedInternal	46	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	47	1	Binary	Number of bitfields to follow.
ReturnBitfield1	48	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfieldn		1	Binary	Last bitfield.
Optional fields. . .				

Example Order Acknowledgment Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`4D 00`	77 bytes
MessageType	`25`	`Order Acknowledgment`
MatchingUnit	`02`	Matching Unit 2
SequenceNumber	`64 00 00 00`	Sequence number 100
TransactionTime	`E0 FA 20 F7 36 71 F8 11`	1,294,909,373,757,320,000
ClOrdID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00`	`ABC123`

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OrderID	05 10 1E B7 5E 39 2F 02	171WC1000005 (base 36)
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	03	3 bitfields to follow
ReturnBitfield1	00	No bitfields from byte 1
ReturnBitfield2	01	*Symbol*
ReturnBitfield3	05	*Account, ClearingAccount*
Symbol	31 32 33 61 42 63 00 00	123aBc
Account	41 42 43 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC
ClearingAccount	00 00 00 00	(empty)

Example Minimal Order Acknowledgment Message:

Field Name	**Hexadecimal**	**Notes**
StartOfMessage	BA BA	Start of message bytes.
MessageLength	2E 00	46 bytes
MessageType	25	`Order Acknowledgment`
MatchingUnit	02	Matching Unit 2
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,320,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
OrderID	05 10 1E B7 5E 39 2F 02	171WC1000005 (base 36)
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	00	No bitfields to follow

4.2.2 Quote Update Acknowledgment

`Quote Update Acknowledgment` messages are sent in response to a `Quote Update` message. The effect of each requested update will be found in this response. The ordering between request and response is preserved.

For quotes which are priced at an executable price and which may remove liquidity, *QuoteResult* reason of "D" or "d" will be provided. In these cases, executions or cancellations (as needed) will immediately follow as additional messages. In some cases, an execution may not be permitted (e.g., risk management causes cancellation of the targeted order before execution), no additional messages will follow and the quote will post.

In some cases, a new *OrderID* will be assigned for an existing quote. There are currently two situations where this occurs, but others may be added in the future:

1. A quote which has received a large number of quote updates over its life will be assigned a new *OrderID* if receiving an update which would cause a loss in priority.
2. A quote update which has had an update to an attribute other than price or size.

If using the *OrderID* in your system or to correlate with an *OrderID* on PITCH, always be prepared to receive an update on a `Quote Update Acknowledgment`.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x71`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.

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				Unsequenced application message. *MatchingUnit* will be set to 0.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit. This will be an unsequenced application message. The sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	18	16	Text	Echoed back from the Quote Update request.
QuoteRejectReason	34	1	Text	Reason for rejection of an entire `Quote Update` message by the matching engine. If an error is indicated, then no quotes were entered or updated. *QuoteCnt* will be 0. `<space>` = Success See **Quote Reason Codes** for a list of possible quote reject codes. Additional reasons may be added in the future without warning.
Reserved	35	17	Binary	Reserved for future expansion. Filled with 0.
QuoteCnt	52	1	Binary	Number of repeating groups included in this acknowledgment. Allowed values are 1-20.
Repeating Groups of …				
OrderID		8	Binary	Order ID assigned by the matching engine. Corresponds to order ID on PITCH.
QuoteResult		1	Text	Result of the quote request. *Acceptance:* A = New Quote L = Modified; loss of priority R = Modified; retains priority (size reduction) N = No change, matches existing quote D = New Quote, but may remove liquidity d = Modified, but may remove liquidity *Cancellation:* U = User cancelled (zero size/price requested) *Rejection:* a = Admin P = Rejected, can't post f = Risk management EFID or Custom Group ID level S = Rejected, symbol not found p = Rejected, invalid price s = Risk management product root level n = Risk management configuration is not sufficient u = Rejected, other reason Additional reasons indicating a reject may be added in the future with no notice.

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SubLiquidity Indicator		1	Text	**CFE may add additional values without notice. Members must gracefully ignore unknown values** ASCII NULL (0x00) = No Additional Information C = Carried Order Indicator U = Qualifying Market Turner order N = Normal
Reserved		6	Binary	Reserved for future expansion. Filled with 0.

Example Quote Update Acknowledgment Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	53 00	83 bytes
MessageType	71	Quote Update Acknowledgment
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
QuoteUpdateID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00	ABC123
QuoteRejectReason	20	<space> = Success
Reserved	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
QuoteCnt	02	Two Quotes
OrderID	05 10 1E B7 5E 39 2F 02	171WC1000005 (base 36)
QuoteResult	41	A = New Quote
SubLiquidityIndicator	55	U = Qualifying Market Turner order
Reserved	00 00 00 00 00 00	
OrderID	06 10 1E B7 5E 39 2F 02	171WC1000006 (base 36)
QuoteResult	53	S = Rejected, symbol not found
SubLiquidityIndicator	00	(unset)
Reserved	00 00 00 00 00 00	

4.2.3 Order Rejected

`Order Rejected` messages are sent in response to a `New Order` which must be rejected. This message corresponds to a FIX Execution Report with *ExecType* (150) = 8 (Rejected). `Order Rejected` messages are unsequenced.

Permitted return optional fields are described in 'Section 6.2 – Order Rejected'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	`0x26`
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application message. Sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).

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ClOrdID	18	20	Text	Echoed back from the original order.
OrderRejectReason	38	1	Text	Reason for an order rejection. See 'Section 8 – Reason Codes' for a list of possible reasons.
Text	39	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	99	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	100	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	101	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfield$_n$		1	Binary	Last bitfield.
Optional fields. . .				

Example Order Rejected Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes
MessageLength	7A 00	122 bytes
MessageType	26	Order Rejected
MatchingUnit	00	Unsequenced Message, unit = 0
SequenceNumber	00 00 00 00	Unsequenced Message, sequence = 0
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,320,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
OrderRejectReason	44	D
Text	44 75 70 6C 69 63 61 74 65 20 43 6C 4F 72 64 49 44 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	Duplicate ClOrdID
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	04	4 bitfields to follow
ReturnBitfield1	00	No bitfields from byte 1
ReturnBitfield2	01	*Symbol*
ReturnBitfield3	06	*ClearingFirm, ClearingAccount*
Symbol	31 32 33 61 42 63 00 00	123aBc
ClearingFirm	54 45 53 54	TEST
ClearingAccount	00 00 00 00	(empty)
MaturityDate	F0 C5 33 01	2/24/2017

4.2.4 Quote Update Rejected

`Quote Update Rejected` messages are sent in response to a `Quote Update` message when the entire quote block is rejected by the order handler. No existing quotes are updated or cancelled as a result.

Field	Offset	Length	Data Type	Description

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StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x78`
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application message. Sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	18	16	Text	Echoed back from the `Quote Update` request.
QuoteRejectReason	34	1	Text	Reason for rejection of an entire `Quote Update` message. See **Quote Reason Codes** for a list of possible quote reject codes. Additional reasons may be added in the future without warning.
Reserved	35	17	Binary	Reserved for future expansion. Filled with 0.

Example Quote Update Rejected Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes
MessageLength	`32 00`	50 bytes
MessageType	`78`	`Quote Update Rejected`
MatchingUnit	`00`	Unsequenced message, unit = 0
SequenceNumber	`00 00 00 00`	Unsequenced message, sequence = 0
TransactionTime	`E0 FA 20 F7 36 71 F8 11`	1,294,909,373,757,324,000
QuoteUpdateID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00`	ABC123
QuoteRejectReason	`4D`	M = symbols not on same matching engine
Reserved	`00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00`	Reserved

4.2.5 Order Modified

`Order Modified` messages are sent in response to a `Modify Request` to indicate that the order has been successfully modified.

Note: You must opt-in to receiving *LeavesQty* in `Order Modified` messages. In some cases, the last message to be received on an order's lifecycle will be an `Order Modified` message. The way to know the order is no longer live is to inspect *LeavesQty*. An example of this would be modification of an order whilst an execution is being generated, resulting in the order being reduced to zero outstanding quantity.

Permitted return optional fields are described in 'Section 6.3 – Order Modified'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	`0x27`

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MatchingUnit	5	1	Binary	The Matching Unit which created this message. Matching units in BOE correspond to Matching Units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per Matching Unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent)
ClOrdID	18	20	Text	Client order ID. This is the *ClOrdID* from the Modify Order message.
OrderID	38	8	Binary	Corresponds to *OrderID* (37) in CFE FIX. The unique *OrderID*. Modifications do not change the *OrderID*.
ReservedInternal	46	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	47	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	48	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfield$_n$		1	Binary	Last bitfield.
Optional fields. . .				

Example Order modified Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	3F 00	63 bytes
MessageType	27	Order Modified
MatchingUnit	02	Matching Unit 2
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
OrderID	05 10 1E B7 5E 39 2F 02	171WC1000005 (base 36)
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	05	5 bitfields to follow
ReturnBitfield1	04	*Price*
ReturnBitfield2	00	No fields from byte 2
ReturnBitfield3	00	No fields from byte 3
ReturnBitfield4	00	No fields from byte 4
ReturnBitfield5	02	*LeavesQty*
Price	08 E2 01 00 00 00 00 00	12.34
LeavesQty	00 00 00 00	0 (order done)

4.2.6 Quote Restated

`Quote Restated` messages are sent to inform the TPH that an order has been asynchronously modified for some reason by the Exchange. Additional reasons may be added in the future.

In the case where an inbound quote will execute against a resting order or quote, then a `Quote Restated` message will be sent after the `Quote Update Acknowledgement` as a function of normal system behavior. These Restatements will contain the *RestatementReason* of "Q = Liquidity".

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This message may be expanded in length in the future with new fields added to the end. To maintain forward compatibility, be prepared to receive a message longer than the documented length and to gracefully ignore those extra fields.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x72`
MatchingUnit	5	1	Binary	The Matching Unit which created this message. Matching units in BOE correspond to Matching Units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per Matching Unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	18	16	Text	Echoed back from the most recent `Quote Update` request for this quote.
OrderID	34	8	Binary	Corresponds to *OrderID* (37) in CFE FIX. The unique *OrderID*. For informational purposes only. Restatements do *not* change the *OrderID*.
LeavesQty	42	4	Binary	New quantity available for execution
WorkingPrice	46	8	Binary	New working price
Symbol	54	6	Alphanumeric	CFE native identifier
Side	60	1	Alphanumeric	`1` = Buy `2` = Sell
RestatementReason	61	1	Alphanumeric	The reason for this Quote Restated message. `Q` = Liquidity `W` = Wash CFE reserves the right to add new values as necessary without prior notice.

Example Quote Restated Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`3C 00`	60 bytes
MessageType	`72`	`Quote Restated`
MatchingUnit	`02`	Matching Unit 2
SequenceNumber	`64 00 00 00`	Sequence number 100
TransactTime	`E0 FA 20 F7 36 71 F8 11`	1,294,909,373,757,324,000
QuoteUpdateID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00`	ABC123
OrderID	`05 10 1E B7 5E 39 2F 02`	171WC1000005 (base 36)
LeavesQty	`14 00 00 00`	20 contracts
WorkingPrice	`AC 07 01 00 00 00 00 00`	6.75
Symbol	`30 30 34 63 53 73`	004cSs
Side	`31`	1 = Buy
RestatementReason	`51`	Q = Liquidity

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4.2.7 User Modify Rejected

`User Modify Rejected` messages are sent in response to a `Modify Order` for an order which cannot be modified. `User Modify Rejected` messages are unsequenced.

This message corresponds to a FIX Execution Report with *MsgType* (35) = 9 (Order Cancel Reject) and *CxlRejResponseTo* (434) = 2 (Order Cancel/Replace Request).

Permitted return optional fields are described in 'Section 6.4 – User Modify Rejected'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x29`
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application message. Sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	18	20	Text	The *ClOrdID* of the modify request which was rejected.
ModifyReject Reason	38	1	Text	Reason for a modify rejection. See 'Section 8 – Reason Codes' for a list of possible reasons.
Text	39	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	99	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	100	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	101	1	Binary	Bitfield identifying fields to return.
…				
$ReturnBitfield_n$		1	Binary	Last bitfield.
Optional fields. . .				

Example User Modify Rejected Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	63 00	99 bytes
MessageType	29	User Modify Rejected
MatchingUnit	00	Unsequenced Message, unit = 0
SequenceNumber	00 00 00 00	Unsequenced Message, sequence = 0
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
ModifyRejectReason	50	Pending Fill
Text	50 65 6E 64 69 6E 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	Pending

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	00 00 00 00 00 00 00 00 00 00	
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	00	No optional fields

4.2.8 Order Cancelled

An order has been cancelled.

Permitted return optional fields are described in 'Section 6.5 – Order Cancelled'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	0x2A
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	18	20	Text	The order which was cancelled.
CancelReason	38	1	Text	Reason for the order cancellation. See 'Section 8 – Reason Codes' for a list of possible reasons.
ReservedInternal	39	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	40	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	41	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfield$_n$		1	Binary	Last bitfield.
Optional fields. . .				

Example Order Cancelled Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes
MessageLength	48 00	72 bytes
MessageType	2A	Order Cancelled
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
CancelReason	55	U = User Requested
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	05	5 bitfields to follow
ReturnBitfield1	00	No fields from byte 1
ReturnBitfield2	00	No fields from byte 2

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ReturnBitfield3	06	*ClearingFirm*, *ClearingAccount*
ReturnBitfield4	00	No fields from byte 2
ReturnBitfield5	01	*OrigClOrdID*
ClearingFirm	54 45 53 54	TEST
ClearingAccount	31 32 33 34	1234
OrigClOrdID	41 42 43 31 32 31 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC121

4.2.9 Quote Cancelled

A `Quote Cancelled` message will be sent to indicate an unsolicited cancellation of a quote entered with a `Quote Update` message. An unsolicited cancellation is used, for example, when a resting quote is cancelled due to MTP with an inbound order or quotes are being cancelled due to a risk trip.

This message may be expanded in length in the future with new fields added to the end. To maintain forward compatibility, be prepared to receive a message longer than the documented length and to gracefully ignore those extra fields.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x73`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH. Unsequenced application message. *MatchingUnit* will be set to 0.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit. This will be an unsequenced application message. The sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	18	16	Text	Echoed back from the most recent `Quote Update` request for this quote.
OrderID	34	8	Binary	Order ID assigned by the matching engine
Symbol	42	6	Alphanumeric	CFE native identifier
Side	48	1	Alphanumeric	1 = Buy 2 = Sell
CancelReason	49	1	Text	Reason for the quote cancellation. See **Order Reason Codes** for a list of possible reasons.

Example Quote Cancelled Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	30 00	48 bytes
MessageType	73	`Quote Cancelled`
MatchingUnit	01	Matching Unit 1

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SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
QuoteUpdateID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00	ABC123
OrderID	05 10 1E B7 5E 39 2F 02	171WC1000005 (base 36)
Symbol	30 30 36 69 70 41	006ipA
Side	32	2 = Sell
CancelReason	41	A = Admin

4.2.10 Cancel Rejected

A `Cancel Rejected` message is sent in response to a `Cancel Order` message to indicate that the cancellation cannot occur. `Cancel Rejected` messages are unsequenced.

Permitted return bitfields are described in 'Section 6.6 - Cancel Rejected'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x2B`
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application message. Sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	18	20	Text	The order whose cancel was rejected.
CancelRejectReason	38	1	Text	See 'Section 8 – Reason Codes' for a list of possible reasons.
Text	39	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	99	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	100	1	Binary	Number of bitfields to follow.
$ReturnBitfield_1$	101	1	Binary	Bitfield identifying fields to return.
$ReturnBitfield_n$		1	Binary	Last bitfield.
Optional fields. . .				

Example Cancel Rejected Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes
MessageLength	63 00	99 bytes
MessageType	2B	`Cancel Rejected`
MatchingUnit	00	Unsequenced Message, unit = 0
SequenceNumber	00 00 00 00	Unsequenced Message, sequence = 0
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
CancelRejectReason	4A	J

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Text	54 4F 4F 20 4C 41 54 45 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	TOO LATE
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	00	No optional fields

4.2.11 Order Execution

An `Order Execution` is sent for each fill on an order.

Rather than returning a monetary value indicating the rebate or charge for an execution, the *FeeCode* is an indication of a fee classification corresponding to an item on the venue's fee schedule.

Permitted return bitfields are described in 'Section 6.7 – Order Execution'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	`0x2C`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	*6*	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	*10*	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	*18*	20	Text	Order receiving the execution.
ExecID	*38*	8	Binary	Corresponds to *ExecID* (17) in CFE FIX. Sent to the OCC in the Trade ID field. Execution ID. Unique across all matching units on a given day. Note: *ExecIDs* will be represented on ODROP and FIXDROP ports as base 36 ASCII. Example conversion: **Decimal** / **Base 36** 28294005440239 / A1234B567 76335905726621 / R248BC23H 728557228187 / 09AP05V2Z
LastShares	*46*	4	Binary	Corresponds to *LastShares* (32) in CFE FIX. Executed share quantity.
LastPx	*50*	8	Binary Price	Corresponds to *LastPx* (31) in CFE FIX. Price of this fill. Note the use *of Binary Price* type to represent positive and negative prices, which can occur with spread instruments.
LeavesQty	*58*	4	Binary	Corresponds to *LeavesQty* (151) in CFE FIX. Quantity still open for further execution. If zero, the order is complete.

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BaseLiquidity Indicator	*62*	1	Alphanumeric	Indicates whether the trade added or removed liquidity. A = Added Liquidity R = Removed Liquidity C = Market opening / re-opening trade
SubLiquidityIndicator	*63*	1	Alphanumeric	ASCII NULL (0x00) U = Qualifying Market Turner order. Only sent when *BaseLiquidityIndicator* = A. **CFE may add additional values without notice. TPHs must gracefully ignore unknown values.**
ContraBroker	64	4	Alphanumeric	Corresponds to *ContraBroker* (375) in CFE FIX. Value always set to "CFE"
ReservedInternal	68	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	69	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	70	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfield$_n$		1	Binary	Last bitfield.
Optional fields…				

Example Order Execution Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes
MessageLength	53 00	83 bytes
MessageType	2C	Order Execution
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
ExecID	01 F0 B7 D9 71 21 00 00	D19800001 (base 36)
LastShares	64 00 00 00	100 contracts
LastPx	08 E2 01 00 00 00 00 00	12.34
LeavesQty	14 00 00 00	20
BaseLiquidityIndicator	41	A = Added
SubLiquidityIndicator	00	(unset)
ContraBroker	43 46 45 00	CFE
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	03	3 bitfields to follow
ReturnBitfield1	00	No bitfields from byte 1
ReturnBitfield2	00	No bitfields from byte 2
ReturnBitfield3	46	*ClearingFirm, ClearingAccount, OrderQty*
ClearingFirm	54 45 53 54	TEST
ClearingAccount	31 32 33 43	1234
OrderQty	78 00 00 00	120 contracts

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4.2.12 Quote Execution

A `Quote Execution` message is used to indicate an execution has occurred on a resting quote.

This message may be expanded in length in the future with new fields added to the end. To maintain forward compatibility, be prepared to receive a message longer than the documented length and to gracefully ignore those extra fields.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x74`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	18	16	Text	Echoed back from the most recent `Quote Update` request for this quote.
OrderID	34	8	Binary	Order ID assigned by the matching engine
ExecID	42	8	Binary	Corresponds to *ExecID* (17) in CFE FIX. Execution ID. Unique across all matching units on a given day. Note: *ExecIDs* will be represented on FIXDROP ports as nine character, base 36 ASCII. Leading zeros should be added if the converted base 36 value is shorter than nine characters. Example conversion: **Decimal** / **Base 36** 28294005440239 / A1234B567 76335905726621 / R248BC23H 728557228187 / 09AP05V2Z
Symbol	50	6	Alphanumeric	CFE native identifier
ClearingFirm	56	4	Alpha	Echoed back from the original quote
LastShares	60	4	Binary	Coresponds to *LastShares* (32) in CFE FIX. Number of contracts being traded.
LastPx	64	8	Binary Price	Corresponds to *LastPx* (31) in CFE FIX. Price of this fill.
LeavesQty	72	4	Binary	Corresponds to *LeavesQty* (151) in CFE FIX. Quantity still open for further execution. If zero, the order is complete.
Side	76	1	Alpha	1 = Buy 2 = Sell
BaseLiquidity Indicator	77	1	Alpha	Indicates whether the trade added or removed liquidity. A = Added Liquidity R = Removed Liquidity C = Market opening / re-opening trade

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SubLiquidityIndicator	78	1	Alpha	**CFE may add additional values without notice. Members must gracefully ignore unknown values.** ASCII NULL (0x00) = No Additional Information C = Carried Order Indicator U = Qualifying Market Turner order
FeeCode	79	2	Alphanumeric	Corresponds to *FeeCode* (9882) in CFE FIX.

Example Quote Execution Message:

Field Name	Hexadecimal	Notes
StartOfMessage	`BA BA`	Start of message bytes.
MessageLength	`56 00`	79 bytes
MessageType	`74`	`Quote Execution`
MatchingUnit	`01`	Matching Unit 1
SequenceNumber	`64 00 00 00`	Sequence number 100
TransactionTime	`E0 FA 20 F7 36 71 F8 11`	1,294,909,373,757,324,000
QuoteUpdateID	`41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00`	ABC123
OrderID	`05 10 1E B7 5E 39 2F 02`	171WC1000005 (base 36)
ExecID	`01 F0 B7 D9 71 21 00 00`	D19800001 (base 36)
Symbol	`30 30 36 69 70 41`	006ipA
ClearingFirm	`41 42 43 44`	ABCD
LastShares	`64 00 00 00`	100 contracts
LastPx	`70 17 00 00 00 00 00 00`	0.60
LeavesQty	`00 00 00 00`	0 (order done)
Side	`31`	1 = Buy
BaseLiquidity Indicator	`41`	A = Added
SubLiquidityIndicator	`55`	U = Qualifying Market Turner Order
FeeCode	`41 42`	AB

4.2.13 Trade Cancel or Correct

Used to relay a trade which has been cancelled (busted) or corrected (price or size change only). The *CorrectedPrice* and optional *CorrectedSize* fields will be set to 0 for cancelled trades and to the new trade price and/or size for corrected trades. `Trade Cancel or Correct` can be sent for same day as well as previous day trades.

Permitted return bitfields are described in 'Section 6.8 – Trade Cancel or Correct'.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x2D`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	18	20	Text	*ClOrdID* of the order whose fill is being cancelled or corrected.

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OrderID	38	8	Binary	Corresponds to *OrderID* (37) in CFE FIX. Order whose fill is being cancelled or corrected.
ExecRefID	46	8	Binary	Corresponds to *ExecRefID* (19) in CFE FIX. Refers to the *ExecID* of the fill being cancelled or corrected.
Side	54	1	Alphanumeric	Side of the order.
BaseLiquidity Indicator	55	1	Alphanumeric	Indicates whether the trade added or removed liquidity. A = Added Liquidity R = Removed Liquidity
ClearingFirm	56	4	Alpha	Echoed back from the original order.
ClearingAccount	60	4	Text	Echoed back from the original order.
LastShares	64	4	Binary	Number of shares of the trade being cancelled.
LastPx	68	8	Binary Price	Price of the trade being cancelled. Note the use of *Binary Price* type to represent positive and negative prices, which can occur with spread instruments.
CorrectedPrice	76	8	Binary Price	For trade corrections, this is the new trade price. For trade breaks, this is set to 0.
OrigTime	84	8	DateTime	Corresponds to *OrigTime* (42). The date and time of the original trade, in GMT.
ReservedInternal	92	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	93	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	94	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfield$_n$		1	Binary	Last bitfield.
Optional fields. . .				

Example Trade Cancel or Correct Message:

Field Name	**Hexadecimal**	**Notes**
StartOfMessage	BA BA	Start of message bytes.
MessageLength	6C 00	108 bytes
MessageType	2D	Trade Cancel or Correct
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
OrderID	05 10 1E B7 5E 39 2F 02	171WC1000005 (base 36)
ExecRefID	01 F0 B7 D9 71 21 00 00	D19800001 (base 36)
Side	31	Buy
BaseLiquidity Indicator	41	A = Added
ClearingFirm	54 45 53 54	TEST
ClearingAccount	00 00 00 00	(empty)
LastShares	64 00 00 00	100 contracts
LastPx	70 17 00 00 00 00 00 00	0.60
CorrectedPrice	00 00 00 00 00 00 00 00	0 (cancelled)
OrigTime	E0 BA 75 95 15 4C EB 11	1,291,209,373,757,324,000
ReservedInternal	00	Ignore

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NumberOfReturn Bitfields	04	4 bitfields to follow
ReturnBitfield1	00	No fields from byte 1
ReturnBitfield2	01	*Symbol*
ReturnBitfield3	00	No fields from byte 3
ReturnBitfield4	01	*MaturityDate*
Symbol	30 30 51 30 6B 41 00 00	00Q0kA
MaturityDate	F0 C5 33 01	2/24/2017

4.2.14 Purge Rejected

A `Purge Rejected` message is sent in response to a `Purge Orders` message to indicate that the mass cancellation cannot occur. `Purge Rejected` messages are unsequenced.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x48`
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application message. Sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
PurgeRejectReason	18	1	Text	Reason for a purge rejection. See 'Section 8 – Reason Codes' for a list of possible reasons.
Text	19	60	Text	Human readable text with more information about the reject reason.
ReservedInternal	79	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	80	1	Binary	Number of bitfields to follow.
ReturnBitfield₁	81	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfieldₙ		1	Binary	Last bitfield.
Optional fields. . .				

Example Purge Rejected Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	72 00	114 bytes
MessageType	48	Purge Rejected
MatchingUnit	00	Unsequenced Message, unit = 0
SequenceNumber	00 00 00 00	Unsequenced Message, sequence = 0
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
PurgeRejectReason	41	A

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Text	41 44 4D 49 4E 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ADMIN
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	0F	15 bitfields to follow
ReturnBitfield1	00	No fields from byte 1
ReturnBitfield2	00	No fields from byte 2
ReturnBitfield3	00	No fields from byte 3
ReturnBitfield4	00	No fields from byte 4
ReturnBitfield5	00	No fields from byte 5
ReturnBitfield6	00	No fields from byte 6
ReturnBitfield7	00	No fields from byte 7
ReturnBitfield8	00	No fields from byte 8
ReturnBitfield9	00	No fields from byte 9
ReturnBitfield10	00	No fields from byte 10
ReturnBitfield11	00	No fields from byte 11
ReturnBitfield12	00	No fields from byte 12
ReturnBitfield13	00	No fields from byte 13
ReturnBitfield14	00	No fields from byte 14
ReturnBitfield15	08	*MassCancelID*
MassCancelID	54 45 53 54 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	TEST

4.2.15 Reset Risk Acknowledgment

Response to a `Reset Risk` request.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	`0x57`
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application message. Sequence number will be set to 0.
RiskStatusID	10	16	Text	Unique identifier for this Reset Risk request. Response message will have this corresponding identifier.
RiskResetResult	26	1	Text	`<space>` = Ignored; exceeds 1 reset per second Y = Success F = Rejected; exceeds firm reset limit C = Rejected; exceeds Custom Group ID limit E = Rejected; empty *ResetRisk* field I = Rejected; Incorrect data center S = Rejected; exceeds product level reset limit U = Rejected; invalid *RiskRoot* c = Rejected; invalid EFID/*ClearingFirm*

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				y = Rejected; in replay Additional reject values may be added in the future with no notice.

Example Risk Reset Acknowledgment Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	19 00	25 bytes
MessageType	57	Risk Reset Acknowledgement
MatchingUnit	00	Unsequenced Message, unit = 0
SequenceNumber	00 00 00 00	Unsequenced Message, sequence = 0
RiskStatusID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00	ABC123
RiskResetResult	59	Y = Success

4.2.16 Mass Cancel Acknowledgment

A Mass Cancel Acknowledgment is an unsequenced message sent when a Cancel Order or Purge Orders message requesting a mass cancellation has completed cancelling all individual orders.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the *StartOfMessage* field.
MessageType	4	1	Binary	0x36
MatchingUnit	5	1	Binary	Unsequenced application message. Matching unit will be set to 0.
SequenceNumber	6	4	Binary	Unsequenced application. Message. Sequence number will be set to 0.
TransactionTime	10	8	DateTime	The time in the order entry gateway when the final matching engine event was received to complete the mass cancel.
MassCancelID	18	20	Text	Copied from the MassCancelID passed on the original Cancel Order. This field corresponds to *MassCancelID* (7695) in CFE FIX.
CancelledOrder Count	38	4	Binary	Number of orders cancelled. This field corresponds to *CancelledOrderCount* (7696) in CFE FIX.
ReservedInternal	42	1	Binary	Reserved for CFE internal use.

Example Mass Cancel Acknowledgment Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA	Start of message bytes.
MessageLength	29 00	41 bytes
MessageType	36	Mass Cancel Acknowledgment
MatchingUnit	00	Unsequenced Message, unit = 0
SequenceNumber	00 00 00 00	Unsequenced Message, sequence = 0
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
MassCancelID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
CancelledOrderCount	63 00 00 00	99 orders were cancelled

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ReservedInternal	00	Ignore

4.2.17 TAS Restatement (New Order Message)

A TAS Restatement is sent post-settlement time for each TAS (VXT) execution during the associated business day to communicate the updated Price and Symbol associated with the cleared execution. TAS Restatement messages are sent shortly after the VX contract settlement prices are disseminated (shortly after 3:15 p.m. CT).

TPHs that trade TAS should register for at least the following fields on TAS Restatement messages.

- *ClearingPrice* – Field contains the as-traded price (*LastPx*) offset with the underlying contract settlement price. This is the price used for TAS trade clearing.
- *ClearingSymbol* – Field contains the mapped symbol ID of the VX contract assocated with the traded VXT contract with the same expiration. This is the symbol under which the TAS trade clears.

See 'Section 6.10 – TAS Restatement' for a complete specification of available fields for TAS Restatement messages.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be 0xBA 0xBA.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	0x49
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	*6*	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	*10*	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	*18*	20	Text	Copied from the associated VXT execution.
ExecID	*38*	8	Binary	Copied from the associated VXT execution.
ReservedInternal	46	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	47	1	Binary	Number of bitfields to follow.
ReturnBitfield$_1$	48	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfield$_n$		1	Binary	Last bitfield.
Optional fields…				

Example TAS Restatement Message (Negative Price):

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	46 00	70 bytes
MessageType	49	TAS Restatement
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
ExecID	01 F0 B7 D9 71 21 00 00	D19800001 (base 36)
ReservedInternal	00	Ignore

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NumberOfReturn Bitfields	0C	12 bitfields to follow
ReturnBitfield₁	00	No bitfields from byte 1
ReturnBitfield₂	01	*Symbol*
…		
ReturnBitfield₁₁	00	No bitfields from byte 11
ReturnBitfield₁₂	50	*ClearingSymbol, ClearingPrice*
Symbol	31 32 33 61 62 63	123abc
ClearingSymbol	34 35 36 64 65 66	456def
ClearingPrice	54 4A 02 00 00 00 00 00	15.01

4.2.18 Variance Restatement (New Order Message)

A `Variance Restatement` is sent post-settlement time for each VA and VAO execution during the associated business day is used to communicate updated Price, Size and Symbol associated with the cleared execution. `Variance Restatement` messages are sent shortly after the S&P 500 index settlement price is received (4:00 p.m. CT).

TPHs that trade Variance futures should register for at least the following fields on `Variance Restatement` messages.

- *ClearingPrice* – *Price* converted to Variance units.
- *ClearingSize* – For VA executions, this field will contain the *LastShares* of the original execution in Vega contracts to Variance units. For VAO executions, this field will contain a copy of the *LastShares* from the original execution as VAO trades directly in Variance units.
- *ClearingSymbol* – Field contains the mapped symbol ID of the VA contract assocated with the traded contract with the same expiration. This is the symbol under which the Variance trade clears. Note that this field will be the same as the as-traded original symbol for VA executions; only VAO executions will experience a change of symbol for clearing.

See 'Section 6.11 – Variance Restatement' for a complete specification of available fields for `Variance Restatement` messages.

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	0x4A
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	*6*	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	*10*	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
ClOrdID	*18*	20	Text	Copied from the associated VA/VAO execution.
ExecID	*38*	8	Binary	Copied from the associated VA/VAO execution.
ReservedInternal	46	1	Binary	Reserved for CFE internal use.
NumberOfReturn Bitfields	47	1	Binary	Number of bitfields to follow.
ReturnBitfield₁	48	1	Binary	Bitfield identifying fields to return.
…				
ReturnBitfieldₙ		1	Binary	Last bitfield.

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Optional fields…				

Example Variance Restatement Message:

Field Name	Hexadecimal	Notes
StartOfMessage	BA BA	Start of message bytes.
MessageLength	4A 00	74 bytes
MessageType	4A	Variance Restatement
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
ClOrdID	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00 00 00 00 00	ABC123
ExecID	01 F0 B7 D9 71 21 00 00	D19800001 (base 36)
ReservedInternal	00	Ignore
NumberOfReturn Bitfields	12	12 bitfields to follow
ReturnBitfield$_1$	00	No bitfields from byte 1
ReturnBitfield$_2$	01	*Symbol*
…		
ReturnBitfield$_{11}$	00	No bitfields from byte 11
ReturnBitfield$_{12}$	70	*ClearingPrice, ClearingSize, ClearingSymbol*
Symbol	31 32 33 61 62 63	123abc
ClearingPrice	00 10 27 00 00 00 00 00	256.00
ClearingSize	0C 1C 00 00	3100
ClearingSymbol	34 35 36 64 65 66	123abc

4.2.19 TAS Quote Restatement

A `TAS Quote Restatement` is sent post-settlement time for each TAS (VXT) quote execution during the associated business day to communicate the updated Price and Symbol associated with the cleared execution. `TAS Quote Restatement` messages are sent shortly after the VX contract settlement prices are disseminated (shortly after 3:15 p.m. CT).

Field	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	`0x75`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	6	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	10	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	18	16	Text	Copied from the Quote Update.
ExecID	34	8	Binary	Copied from the associated VXT execution.
Symbol	42	6	Alphanumeric	CFE native identifier
ClearingSymbol	48	8	Alphanumeric	Corresponds to *ClearingSymbol* (21053) in CFE FIX.

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				Optional field used in restatement messages where the symbol on which the original execution occurred is tranformed prior to clearing.
ClearingPrice	56	8	Binary Price	Corresponds to *ClearingPrice* (21050) in CFE FIX. Optional field used in restatement messages where the originally reported fill price (*LastPx*) is transformed prior to clearing.
Reserved	64	16	Binary	Reserved for future expansion. To maintain forward compatibility, fill with 0.

Example TAS Restatement Message (Negative Price):

Field Name	**Hexadecimal**	**Notes**
StartOfMessage	BA BA	Start of message bytes.
MessageLength	4E 00	78 bytes
MessageType	75	TAS Restatement
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
QuoteUpdateId	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00	ABC123
ExecID	01 F0 B7 D9 71 21 00 00	D19800001 (base 36)
Symbol	31 32 33 61 62 63	123abc
ClearingSymbol	34 35 36 64 65 66 00 00	456def
ClearingPrice	54 4A 02 00 00 00 00 00	15.01
Reserved	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	Reserved

4.2.20 Variance Quote Restatement

A `Variance Quote Restatement` is sent post-settlement time for each VA and VAO execution during the associated business day is used to communicate updated Price, Size and Symbol associated with the cleared execution. `Variance Restatement` messages are sent shortly after the S&P 500 index settlement price is received (4:00 p.m. CT).

Field	**Offset**	**Length**	**Data Type**	**Description**
StartOfMessage	0	2	Binary	Must be `0xBA 0xBA`.
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes for the StartOfMessage field.
MessageType	4	1	Binary	`0x76`
MatchingUnit	5	1	Binary	The matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH.
SequenceNumber	*6*	4	Binary	The sequence number for this message. Distinct per matching unit.
TransactionTime	*10*	8	DateTime	The time the event occurred in the CFE Matching Engine (not the time the message was sent).
QuoteUpdateID	*18*	16	Text	Copied from the Quote Update.
ExecID	*34*	8	Binary	Copied from the associated VXT execution.

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Symbol	42	6	Alphanumeric	CFE native identifier
ClearingSymbol	48	8	Alphanumeric	Corresponds to *ClearingSymbol* (21053) in CFE FIX. Optional field used in restatement messages where the symbol on which the original execution occurred is tranformed prior to clearing.
ClearingPrice	56	8	Binary Price	Corresponds to *ClearingPrice* (21050) in CFE FIX. Optional field used in restatement messages where the originally reported fill price (*LastPx*) is transformed prior to clearing.
ClearingSize	64	4	Binary	Corresponds to *ClearingSize* (21051) in CFE FIX. Optional field used in `Order Execution` messsages corresponding to VA and VAO trades, and `Variance Restatement` messages where the originally reported fill quantity (*LastShares*) is transformed prior to clearing.
Reserved	68	16	Binary	Reserved for future expansion. To maintain forward compatibility, fill with 0.

Example Variance Restatement Message:

Field Name	**Hexadecimal**	**Notes**
StartOfMessage	BA BA	Start of message bytes.
MessageLength	52 00	82 bytes
MessageType	76	Variance Restatement
MatchingUnit	01	Matching Unit 1
SequenceNumber	64 00 00 00	Sequence number 100
TransactionTime	E0 FA 20 F7 36 71 F8 11	1,294,909,373,757,324,000
QuoteUpdateId	41 42 43 31 32 33 00 00 00 00 00 00 00 00 00 00	ABC123
ExecID	01 F0 B7 D9 71 21 00 00	D19800001 (base 36)
Symbol	31 32 33 61 62 63	123abc
ClearingSymbol	34 35 36 64 65 66 00 00	456def
ClearingPrice	54 4A 02 00 00 00 00 00	15.01
ClearingSize	0C 1C 00 00	3100
Reserved	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	Reserved

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5 Input Bitfields Per Message

Legend:

R Indicates that the field must be specified for a message
● Indicates that the field can be specified for a message
(Blank) Indicates that the field is not used by CFE and cannot be specified for a message

Input messages that containing invalid fields (i.e., Blank) will be rejected. In the case of rejected input messages, the associated `Reject` message sent back to the TPH will contain a 'RejectReason' code non-optional field (See Section 8 – Reason Codes) and a 'Text' non-optional field containing descriptive text.

5.1 New Order

Byte	Bit	Field	
1	1	*ClearingFirm*	●
	2	*ClearingAccount*	●
	4	*Price*	●
	8	*ExecInst*	
	16	*OrdType*	●
	32	*TimeInForce*	**R**
	64	*MinQty*	●
	128	*MaxFloor*	
2	1	*Symbol*	**R**
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	**R**
	128	*RoutingInst*	
3	1	*Account*	**R**
	2	*DisplayIndicator*	
	4	*MaxRemovePct*	
	8	*DiscretionAmount*	
	16	*PegDifference*	
	32	*PreventMatch*	●
	64	*LocateRequired*	
	128	*ExpireTime*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*RiskReset*	●
	16	*OpenClose*	●
	32	*CMTANumber*	●
	64	*TargetPartyID*	
	128	*(Reserved)*	

Byte	Bit	Field	
5	1	*(Reserved)*	
	2	*AttributedQuote*	
	4	*BookingType*	
	8	*ExtExecInst*	
	16	*ClientID*	
	32	*InvestorID*	
	64	*ExecutorID*	
	128	*OrderOrigination*	
6	1	*DisplayRange*	
	2	*StopPx*	●
	4	*RoutStrategy*	
	8	*RouteDeliveryMethod*	
	16	*ExDestination*	
	32	*EchoText*	
	64	*AuctionId*	
	128	*RoutingFirmID*	
7	1	*AlgorithmicIndicator*	
	2	*CustomGroupId*	●
	4	*ClientQualifiedRole*	
	8	*InvestorQualifiedRole*	
	16	*ExecutorQualifiedRole*	
	32	*CtiCode*	**R**
	64	*ManualOrderIndicator*	**R**
	128	*OEOID*	**R**
8	1	*RoomID*	
	2	*SIIndicator*	
	4	*ClearingOptionalData*	
	8	*ClientIdAttr*	
	16	*FrequentTraderID*	●
	32	*Compression*	
	64	*FloorDestination*	
	128	*FloorRoutingInst*	

Byte	Bit	Field	
9	1	*OrderOrigin*	
	2	*ORS*	
	4	*PriceType*	
	8	*TradingSessionId*	
	16	*CustOrderHandlingInst*	●
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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5.2 Cancel Order

Byte	Bit	Field	
1	1	*ClearingFirm*	●
	2	*MassCancelLockout*	
	4	*MassCancel*	
	8	*ProductName*	●
	16	*MassCancelID*	●
	32	*RoutingFirmID*	
	64	*ManualOrderIndicator*	**R**
	128	*OEOID*	**R**
2	1	*MassCancelInst*	●
	2	*(Reserved)*	
	4	*(Reserved)*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

5.3 Modify Order

Byte	Bit	Field	
1	1	*ClearingFirm*	●
	2	*(Reserved)*	
	4	*OrderQty*	**R**
	8	*Price*	**R**
	16	*OrdType*	●
	32	*CancelOrigOnReject*	●
	64	*ExecInst*	
	128	*Side*	
2	1	*MaxFloor*	
	2	*StopPx*	●
	4	*RoutingFirmID*	
	8	*ManualOrderIndicator*	**R**
	16	*OEOID*	**R**
	32	*FrequentTraderID*	●
	64	*CustOrderHandlingInst*	●
	128	*(Reserved)*	

OrderQty, Price, Manual Order Indicator and OEOID must be present on all `Modify Order` requests. Messages sent without these fields will be rejected.

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5.4 Purge Orders

Byte	Bit	Field	
1	1	*ClearingFirm*	●
	2	*MassCancelLockout*	
	4	*MassCancelInst*	**R**
	8	*ProductName*	●
	16	*MassCancelID*	●
	32	*RoutingFirmID*	
	64	*ManualOrderIndicator*	**R**
	128	*OEOID*	**R**
2	1	*Symbol*	
	2	*SymbolSfx*	
	4	*(Reserved)*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6 Return Bitfields Per Message

Legend:

- ● Indicates that the field can be requested for a message
- – Indicates that the field cannot be requested for a message
- (Blank) Indicates that the field is not used by CFE and cannot be requested for a message

Attempts to register to receive an unused field in the `Login Request` message will result in a rejected login.

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6.1 Order Acknowledgment

Byte	Bit	Field	
1	1	Side	●
	2	PegDifference	
	4	Price	●
	8	ExecInst	
	16	OrderType	●
	32	TimeInForce	●
	64	MinQty	●
	128	MaxRemovePct	
2	1	Symbol	●
	2	SymbolSfx	
	4	Currency	
	8	IdSource	
	16	SecurityId	
	32	SecurityExchange	
	64	Capacity	●
	128	(Reserved)	
3	1	Account	●
	2	ClearingFirm	●
	4	ClearingAccount	●
	8	DisplayIndicator	
	16	MaxFloor	
	32	DiscretionAmount	
	64	OrderQty	●
	128	PreventMatch	●
4	1	MaturityDate	●
	2	StrikePrice	
	4	PutOrCall	
	8	OpenClose	●
	16	ClOrdIdBatch	
	32	CorrectedSize	–
	64	PartyID	
	128	AccessFee	
5	1	OrigClOrdId	–
	2	LeavesQty	●
	4	LastShares	–
	8	LastPx	–
	16	DisplayPrice	
	32	WorkingPrice	
	64	BaseLiquidityIndicator	●
	128	ExpireTime	●
6	1	SecondaryOrderId	–
	2	CCP	
	4	ContraCapacity	
	8	AttributedOrder	
	16	ExtExecInst	
	32	BulkOrderIds	
	64	BulkRejectReasons	
	128	PartyRole	
7	1	SubLiquidityIndicator	●
	2	TradeReportTypeReturn	
	4	TradePublishIndReturn	
	8	Text	
	16	Bid	
	32	Offer	
	64	LargeSize	
	128	LastMkt	
8	1	FeeCode	–
	2	EchoText	
	4	StopPx	●
	8	RoutingInst	
	16	RoutStrategy	
	32	RouteDeliveryMethod	
	64	ExDestination	
	128	TradeReportRefID	
9	1	MarketingFeeCode	
	2	TargetPartyID	
	4	AuctionId	
	8	OrderCategory	
	16	LiquidityProvision	
	32	CmtaNumber	●
	64	CrossType	
	128	CrossPrioritization	
10	1	CrossId	
	2	AllocQty	
	4	GiveUpFirmID	
	8	RoutingFIrmID	
	16	WaiverType	
	32	CrossExclusionIndicator	
	64	PriceFormation	
	128	ClientQualifiedRole	
11	1	ClientID	
	2	InvestorID	
	4	ExecutorID	
	8	OrderOrigination	
	16	Algo	
	32	DeferralReason	
	64	InvestorQualifiedRole	
	128	ExecutorQualifiedRole	
12	1	CtiCode	●
	2	ManualOrderIndicator	●
	4	OEOID	●
	8	TradeDate	–
	16	ClearingPrice	–
	32	ClearingSize	–
	64	ClearingSymbol	–
	128	ClearingOptionalData	
13	1	CumQty	●
	2	DayOrderQty	●
	4	DayCumQty	●
	8	AvgPx	●
	16	DayAvgPx	●
	32	PendingStatus	–
	64	DrillThruProtection	
	128	MultilegReportingType	–
14	1	LegCFICode	
	2	LegMaturityDate	
	4	LegStrikePrice	
	8	RoomId	
	16	SecondaryExecId	–
	32	UserRequestID	
	64	SISUsername	
	128	UserStatus	
15	1	TradeReportingIndicator	
	2	EquityPartyId	
	4	EquityNBBOProtect	
	8	MassCancelId	–
	16	TradePublishInd	
	32	ReportTime	
	64	LegSymbolSfx	
	128	ClientIdAttr	
16	1	FrequentTraderId	●
	2	SessionEligibility	
	4	ComboOrder	
	8	Compression	
	16	FloorDestination	
	32	FloorRoutingInst	
	64	MultiClassSprd	
	128	OrderOrigin	
17	1	PriceType	
	2	StrategyID	
	4	TradingSessionId	
	8	TradeThroughAlertType	
	16	SenderLocationID	
	32	FloorTraderAcronym	
	64	ExecLegCFICode	
	128	CustOrderHandlingInst	●
18	1	(Reserved)	
	2	CrossInitiator	
	4	Subreason	
	8	(Reserved)	
	16	(Reserved)	
	32	(Reserved)	
	64	(Reserved)	
	128	(Reserved)	

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6.2 Order Rejected

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	●
	2	*ClearingFirm*	●
	4	*ClearingAccount*	●
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	●
	128	*PreventMatch*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFIrmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	●
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.3 Order Modified

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	●
	2	*ClearingFirm*	●
	4	*ClearingAccount*	●
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	●
	128	*PreventMatch*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	●
	2	*LeavesQty*	●
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	●
	128	*ExpireTime*	●
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFIrmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	●
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.4 User Modify Rejected

Byte	Bit	Field	
1	1	*Side*	–
	2	*PegDifference*	
	4	*Price*	–
	8	*ExecInst*	
	16	*OrderType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	
2	1	*Symbol*	–
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	–
	128	*(Reserved)*	
3	1	*Account*	–
	2	*ClearingFirm*	–
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	–
	128	*PreventMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	–
	4	*StopPx*	–
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	–
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OEOID*	–
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	–
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	–
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.5 Order Cancelled

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	●
	2	*ClearingFirm*	●
	4	*ClearingAccount*	●
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	●
	128	*PreventMatch*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	●
	2	*LeavesQty*	●
	4	*LastShares*	●
	8	*LastPx*	●
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	●
6	1	*SecondaryOrderId*	●
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	●
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.6 Cancel Rejected

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	–
	2	*ClearingFirm*	–
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	–
	128	*PreventMatch*	–
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	●
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	–
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.7 Order Execution

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	●
	2	*ClearingFirm*	●
	4	*ClearingAccount*	●
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	●
	128	*PreventMatch*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	●
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	●
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	●
	16	*ClearingPrice*	–
	32	*ClearingSize*	●
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	●
	2	*DayOrderQty*	●
	4	*DayCumQty*	●
	8	*AvgPx*	●
	16	*DayAvgPx*	●
	32	*PendingStatus*	●
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	●
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	●
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	●
	2	*SessionEligibility*	
	3	*ComboOrder*	
	4	*Compression*	
	5	*FloorDestination*	
	6	*FloorRoutingInst*	
	7	*MultiClassSprd*	
	8	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.8 Trade Cancel or Correct

Byte	Bit	Field	
1	1	*Side*	–
	2	*PegDifference*	
	4	*Price*	–
	8	*ExecInst*	
	16	*OrderType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	
2	1	*Symbol*	•
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	•
	128	*(Reserved)*	
3	1	*Account*	–
	2	*ClearingFirm*	–
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	–
	128	*PreventMatch*	–
4	1	*MaturityDate*	•
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	•
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	•
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	–
	4	*StopPx*	–
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	•
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OEOID*	–
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	–
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	–
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.9 Purge Rejected

Byte	Bit	Field	
1	1	*Side*	–
	2	*PegDifference*	
	4	*Price*	–
	8	*ExecInst*	
	16	*OrderType*	–
	32	*TimeInForce*	–
	64	*MinQty*	–
	128	*MaxRemovePct*	
2	1	*Symbol*	–
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	–
	128	*(Reserved)*	
3	1	*Account*	–
	2	*ClearingFirm*	–
	4	*ClearingAccount*	–
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	–
	128	*PreventMatch*	–
4	1	*MaturityDate*	–
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	–
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	–
	2	*LeavesQty*	–
	4	*LastShares*	–
	8	*LastPx*	–
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	–
	2	*EchoText*	–
	4	*StopPx*	–
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	–
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFIrmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	–
	2	*ManualOrderIndicator*	–
	4	*OEOID*	–
	8	*TradeDate*	–
	16	*ClearingPrice*	–
	32	*ClearingSize*	–
	64	*ClearingSymbol*	–
	128	*(Reserved)*	–

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	–
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	–
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	●
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	–
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	–
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.10 TAS Restatement

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	●
	2	*ClearingFirm*	●
	4	*ClearingAccount*	●
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	–
	128	*PreventMatch*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	●
	2	*LeavesQty*	–
	4	*LastShares*	●
	8	*LastPx*	●
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	●
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	●
	16	*ClearingPrice*	●
	32	*ClearingSize*	–
	64	*ClearingSymbol*	●
	128	*(Reserved)*	

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	●
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	●
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	●
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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6.11 Variance Restatement

Byte	Bit	Field	
1	1	*Side*	●
	2	*PegDifference*	
	4	*Price*	●
	8	*ExecInst*	
	16	*OrderType*	●
	32	*TimeInForce*	●
	64	*MinQty*	●
	128	*MaxRemovePct*	
2	1	*Symbol*	●
	2	*SymbolSfx*	
	4	*Currency*	
	8	*IdSource*	
	16	*SecurityId*	
	32	*SecurityExchange*	
	64	*Capacity*	●
	128	*(Reserved)*	
3	1	*Account*	●
	2	*ClearingFirm*	●
	4	*ClearingAccount*	●
	8	*DisplayIndicator*	
	16	*MaxFloor*	
	32	*DiscretionAmount*	
	64	*OrderQty*	–
	128	*PreventParticipantMatch*	●
4	1	*MaturityDate*	●
	2	*StrikePrice*	
	4	*PutOrCall*	
	8	*OpenClose*	●
	16	*ClOrdIdBatch*	
	32	*CorrectedSize*	–
	64	*PartyID*	
	128	*AccessFee*	
5	1	*OrigClOrdId*	●
	2	*LeavesQty*	–
	4	*LastShares*	●
	8	*LastPx*	●
	16	*DisplayPrice*	
	32	*WorkingPrice*	
	64	*BaseLiquidityIndicator*	–
	128	*ExpireTime*	–
6	1	*SecondaryOrderId*	–
	2	*CCP*	
	4	*ContraCapacity*	
	8	*AttributedOrder*	
	16	*ExtExecInst*	
	32	*BulkOrderIds*	
	64	*BulkRejectReasons*	
	128	*PartyRole*	

Byte	Bit	Field	
7	1	*SubLiquidityIndicator*	–
	2	*TradeReportTypeReturn*	
	4	*TradePublishIndReturn*	
	8	*Text*	
	16	*Bid*	
	32	*Offer*	
	64	*LargeSize*	
	128	*LastMkt*	
8	1	*FeeCode*	●
	2	*EchoText*	
	4	*StopPx*	●
	8	*RoutingInst*	
	16	*RoutStrategy*	
	32	*RouteDeliveryMethod*	
	64	*ExDestination*	
	128	*TradeReportRefID*	
9	1	*MarketingFeeCode*	
	2	*TargetPartyID*	
	4	*AuctionId*	
	8	*OrderCategory*	
	16	*LiquidityProvision*	
	32	*CmtaNumber*	●
	64	*CrossType*	
	128	*CrossPrioritization*	
10	1	*CrossId*	
	2	*AllocQty*	
	4	*GiveUpFirmID*	
	8	*RoutingFirmID*	
	16	*WaiverType*	
	32	*CrossExclusionIndicator*	
	64	*PriceFormation*	
	128	*ClientQualifiedRole*	
11	1	*ClientID*	
	2	*InvestorID*	
	4	*ExecutorID*	
	8	*OrderOrigination*	
	16	*Algo*	
	32	*DeferralReason*	
	64	*InvestorQualifiedRole*	
	128	*ExecutorQualifiedRole*	
12	1	*CtiCode*	●
	2	*ManualOrderIndicator*	●
	4	*OEOID*	●
	8	*TradeDate*	●
	16	*ClearingPrice*	●
	32	*ClearingSize*	●
	64	*ClearingSymbol*	●
	128	*(Reserved)*	

Byte	Bit	Field	
13	1	*CumQty*	–
	2	*DayOrderQty*	–
	4	*DayCumQty*	–
	8	*AvgPx*	–
	16	*DayAvgPx*	–
	32	*PendingStatus*	–
	64	*DrillThruProtection*	
	128	*MultilegReportingType*	●
14	1	*LegCFICode*	
	2	*LegMaturityDate*	
	4	*LegStrikePrice*	
	8	*RoomId*	
	16	*SecondaryExecId*	●
	32	*UserRequestID*	
	64	*SISUsername*	
	128	*UserStatus*	
15	1	*TradeReportingIndicator*	
	2	*EquityPartyId*	
	4	*EquityNBBOProtect*	
	8	*MassCancelId*	–
	16	*TradePublishInd*	
	32	*ReportTime*	
	64	*LegSymbolSfx*	
	128	*ClientIdAttr*	
16	1	*FrequentTraderId*	●
	2	*SessionEligibility*	
	4	*ComboOrder*	
	8	*Compression*	
	16	*FloorDestination*	
	32	*FloorRoutingInst*	
	64	*MultiClassSprd*	
	128	*OrderOrigin*	
17	1	*PriceType*	
	2	*StrategyID*	
	4	*TradingSessionId*	
	8	*TradeThroughAlertType*	
	16	*SenderLocationID*	
	32	*FloorTraderAcronym*	
	64	*ExecLegCFICode*	
	128	*CustOrderHandlingInst*	●
18	1	*(Reserved)*	
	2	*CrossInitiator*	
	4	*Subreason*	
	8	*(Reserved)*	
	16	*(Reserved)*	
	32	*(Reserved)*	
	64	*(Reserved)*	
	128	*(Reserved)*	

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7 List of Optional Fields

The following are descriptions of optional fields which may be sent or received.

Field	Length	Data Type	Description
Account	16	Text	Corresponds to *Account* (1) in CFE FIX. Unique account identifier associated with an order. This field will be reflected back on execution reports associated with this order. The first 10 characters are sent to the OCC in the Account # field. The entire 16 character string will appear in the Optional CM Data field. Valid characters include ASCII 32-126.
AvgPx	8	Binary Price	Corresponds to *AvgPx* (6) in CFE FIX. Average price of executions for this order weighted by trade size. Zero if *CumQty* field is zero or if a leg fill related to a complex execution.
BaseLiquidityIndicator	1	Alphanumeric	Corresponds to *TradeLiquidityIndicator* (9730). Indicates whether the trade added or removed liquidity. A A = Added Liquidity R = Removed Liquidity C = Market opening / re-opening trade
CancelOrigOnReject	1	Alpha	Corresponds to *CancelOrigOnReject* (9619) in CFE FIX. Indicates handling of original order on failure to modify. N = Leave original order alone. Y = Cancel original order if modification fails.
Capacity	1	Alpha	Corresponds to *OrderCapacity* (47) in CFE FIX. C = Customer F = Firm The *Capacity* refers to the OCC account type. A value of "C" denotes an account that clears in the Customer range at OCC. A value of "F" denotes an account that clears in the Clearing Firm range at OCC.
ClearingAccount	4	Text	Corresponds to *OnBehalfOfSubID* (116) and *ClearingAccount* (440) in CFE FIX. Supplemental identifier. Recorded and made available in execution reports. Available via Drop feeds. This field can be blank or filled out with an optional four character string. This field is not sent to the OCC.
ClearingFirm	4	Alpha	Corresponds to *OnBehalfOfCompID* (115) CFE FIX. EFID that will clear the trade. **Port attribute value of 'Default EFID' is used if not provided.** Sent to OCC in Exec Broker field.

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ClearingPrice	8	Binary Price	Corresponds to *ClearingPrice* (21050) in CFE FIX. Optional field used in restatement messages where the originally reported fill price (*LastPx*) is transformed prior to clearing.
ClearingSize	4	Binary	Corresponds to *ClearingSize* (21051) in CFE FIX. Optional field used in `Order Execution` messsages corresponding to VA and VAO trades, and `Variance Restatement` messages where the originally reported fill quantity (*LastShares*) is transformed prior to clearing.
ClearingSymbol	8	Alphanumeric	Corresponds to *ClearingSymbol* (21053) in CFE FIX. Optional field used in restatement messages where the symbol on which the original execution occurred is tranformed prior to clearing.
CMTANumber	4	Binary	Corresponds to *CMTANumber* (439) in CFE FIX. CMTA Number of the firm that will clear the trade. Must be specified for CMTA orders and left unspecified for non-CMTA orders. Sent to the OCC in the CMTA CM# field.
ContraCapacity	1	Alphanumeric	Capacity of the contra for this execution. See *Capacity* for allowed values.
CorrectedSize	4	Binary	Corresponds to *CorrectedSize* (6655) in CFE FIX. Number of shares after trade adjustment.
CtiCode	1	Alphanumeric	Corresponds to *CTICode* (9702) in CFE FIX. Valid values: 1, 2, 3, 4 1 = CTI 1: Transactions initiated and executed by an individual TPH for the TPH's own account, for an account the TPH controls, or for the account in which the TPH has an ownership or financial interest. 2 = CTI 2: Transactions executed for the proprietary account of a clearing TPH or non-clearing TPH. 3 = CTI 3: Transactions where an individual TPH or authorized trader executes for the personal account of another individual TPH, for an account the other individual TPH controls or for an account in which the other individual TPH has an ownership or financial interest. 4 = CTI 4: Any transaction not meeting the definition of CTI 1, 2 or 3. (These should be non-TPH customer transactions).
CumQty	4	Binary	Corresponds to *CumQty* (14) in CFE FIX Cumulative quanity of contracts executed for the order over the life of the order, which may be multiple business days in the case of GTC and GTD orders. Populated for leg fills related to complex executions
CustomGroupID	2	Binary	Corresponds to *CustomGroupID* (7699) in CFE FIX for `New Order` and `Purge Orders` messages. Used to group orders for use in mass cancels where multiple orders can be cancelled by specifying a list of *CustomGroupIDs*.

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CustOrderHandlingInst	1	Alphanumeric	Corresponds to *CustOrderHandlingInst* (1031) in CFE FIX. Execution source code provided during order entry to describe broker service. A default value can be set using the 'Default Customer Order Handling Instruction' port attribute. `W` = Desk (high touch) `Y` = Electronic (default) `C` = Vendor-provided platform, billed by Executing Broker `G` = Sponsored Access via Exchange API or FIX, provided by executing broker `H` = Premium algorithmic trading provider, billed by executing broker `D` = Other, including other-provided screen
DayAvePx	8	Binary Price	Corresponds to *DayAvgPx* (426) in CFE FIX. Applicable to GTC and GTD orders only. Average price per contract of executions on current business date. Zero if *DayCumQty* is zero.
DayCumQty	4	Binary	Corresponds to *DayCumQty* (425) in CFE FIX. Applicable to GTC and GTD orders only. Cumulative quantity of contracts executed for the order during the current business day.
DayOrderQty	4	Binary	Corresponds to *DayOrderQty* (424) in CFE FIX. Applicable to GTC and GTD orders only. Contracts remaning to be filled for the order at the beginning of the current business day (i.e., *OrderQty* – *CumQty* at the end of the previous business day)
ExpireTime	8	DateTime	Corresponds to *ExpireTime* (126) in CFE FIX. Required for *TimeInForce* = 6 orders, specifies the date-time (in UTC) that the order expires.
FeeCode	2	Alphanumeric	Corresponds to *FeeCode* (9882) in CFE FIX. Indicates fee associated with an execution. Fee codes are published in the pricing schedule. New fee codes may be sent with little or no notice. TPHs are encouraged to code their systems to accept unknown fee codes.
FrequentTraderId	6	Alphanumeric	Corresponds to *FrequentTraderId* (21097) in CFE FIX. Supplemental customer identifier used for billing related programs.
LastPx	8	Binary Price	Corresponds to *LastPx* (31) in CFE FIX. Price of this fill. Present on MTP triggered `Order Cancelled` message. Contains the price at which *LastShares* would have matched.
LastShares	4	Binary	Corresponds to *LastShares* (32) in CFE FIX. Quantity of contracts traded on this fill. Present on MTP triggered `Order Cancelled` message. Contains the number of contracts that would have been matched.
LeavesQty	4	Binary	Corresponds to *LeavesQty* (151) in CFE FIX. Quantity still open for further execution. If zero, the order is complete.
ManualOrder	1	Alpha	Corresponds to *ManualOrderIndicator* (1028) in CFE FIX.

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Indicator			Y = Manual order entry N = Automated order entry
MassCancelInst	16	Text	Corresponds to *MassCancelInst* (7700) in CFE FIX. Used for specification of `Purge Orders` functionality and optionally used for specification of Mass Cancel functionality associated with the `Cancel Order` message. At least one character must be provided (Clearing Firm Filter). Contiguous characters must be specified up to total length. Truncated/unspecified characters will default to values indicated (D) below. ***1st Character : Clearing Firm Filter*** A = No filtering by clearing firm relationship is performed. F = All orders that were sent under the clearing relationship specified in *ClearingFirm* optional field. If "F" specified and ClearingFirm not provided, the Mass Cancel or Purge request will be rejected. ***2nd Character : Acknowledgement Style*** M = (D) `Order Cancelled` messages are sent for each cancelled order. If "M" is set and the *MassCancelID* optional field is specified, the *MassCancelID* value is ignored unless used in combination with "S" and/ or "B". S = A single `Mass Cancel Acknowledgement` message is sent once all cancels have been processed. The *MassCancelID* optional field must be specified or the Mass Cancel or Purge Request will be rejected. B = Both individual `Order Cancelled` and `Mass Cancel Acknowledgement` messages will be sent. Also requires *MassCancelID* optional filed to be specified or the Mass Cancel or Purge request will be rejected. ***3rd Character : Lockout Instruction*** N = (D) No lockout L = Lockout until corresponding Risk Reset received. Lockout can be used only with Clearing Firm Filter set to "F", otherwise the Mass Cancel or Purge request will be rejected. Lockout will apply to all `New Order`, `Quote Update`, and `Modify Order` messages for the *ClearingFirm* (and *ProductName* or *CustomGroupIDs*, if specified), regardless of other filtering in the `Purge Orders` or `Cancel Order` message. ***4th Character : Instrument Type Filter*** B = (D) Cancel both Simple and Complex orders S = Cancel Simple orders only C = Cancel Spread orders only ***5th Character : GTC Order Filter*** C = (D) Cancel GTC and GTD orders P = Don't cancel (preserve) GTC and GTD orders If *ProductName* optional field is specified, it must contain a valid futures root symbol (e.g., "VX"), in which case only orders/quotes associated with the specified product will be cancelled. A self-imposed lockout can be released using the *RiskReset* field of the `New Order` message or the `Reset Risk` message. An appropriate reset is required to be sent for each lockout type in order to resume trading. For example, a product-level lockout requires a product-level reset. For more information, refer to the CFE Risk Management Specification.

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MassCancelID	20	Text	Corresponds to *MassCancelID* (7695) in CFE FIX. Copied from the *MassCancelID* on the original `Cancel Order` or `Purge Orders` message
MaturityDate	4	Date	Corresponds to *MaturityMonth* (200) and *MaturityDay* (205) in CFE FIX. When specifying the Symbol for a `New Order` message the user can specify the mapped symbol identifier in the *Symbol* field. Alternatively, the product class (e.g., "VX", "VXT", "VU", etc.) can be supplied for the Symbol field and the MaturityDate field is used to specify the expiration date of the symbol within the specified product class. If a value is provided for *MaturityDate*, the Symbol field must correspond to a valid product or the order will be rejected with reason code C (Unknown Product Name). If an invalid *MaturityDate* is provided, the order will be rejected with reason code B (Unknown Maturity Date).
MinQty	4	Binary	Corresponds to *MinQty* (110) in CFE FIX. Minimum fill quantity for IOC orders. Ignored for other Simple instrument orders. Not supported for Spread instruments. Spread instrument orders with specified *MinQty* will be rejected.
MultilegReportingType	1	Alphanumeric	Corresponds to *MultilegReportingType* (442) in CFE FIX Present on `Order Execution`, `TAS Restatement` and `Variance Restatement` messages representing either Spread orders or Simple orders that are part Spread execution. 1 = Simple instrument execution 2 = Simple instrument execution that is part of a Spread execution 3 = Spread instrument execution
OpenClose	1	Alphanumeric	Corresponds to *OpenClose* (77) in CFE FIX. Indicates status of client position in a trade resulting from the order. O = Open C = Close N = None (same as not present)
OEOID	18	Text	Corresponds to *OEOID* (25004) in CFE FIX. Identifies the Order Entry Operator responsible for this message. Min length 3, max length 18. Values in ASCII range 33-126 except comma, semicolon and pipe are permissible.
OrderQty	4	Binary	Corresponds to *OrderQty* (38) in CFE FIX. Order quantity. System limit is 999,999 contracts.
OrdType	1	Alphanumeric	Corresponds to *OrdType* (40) in CFE FIX. 1 = Market 2 = Limit (default) 4 = Stop Limit Market implies *TimeInForce* of IOC (3). Stop Limit orders must have a *TimeInForce* of DAY (0), GTC (1), or GTD (6).

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OrigClOrdID	20	Text	Corresponds to *OrigClOrdID* (41) in CFE FIX.
PendingStatus	1	Alphanumeric	Field is provided as a convenience to determine whether an Order Execution message is a preliminary notification representing a pending trade. The value 'P' indicates that the execution is associated with a product for which the Order Execution message is a preliminary notification of an execution and for which a post-settlement restatement will be sent. N = Not applicable P = Pending See 'Section **Error! Reference source not found.** – **Error! Reference source not found.**' For a description products for which PendingStatus is applicable.
PreventMatch	3	Alpha	Corresponds to *PreventMatch* (7928) in CFE FIX. Three characters: ***1st character – MTP Modifier:*** N = Cancel Newest O = Cancel Oldest B = Cancel Both ***2nd character – Unique ID Level:*** F = Prevent Match at Firm(TPH) Level M = Prevent Match at EFID Level ***3rd character – Trading Group ID (optional):*** TPH specified alphanumeric value 0-9, A-Z, or a-z. The Unique ID level (character 2) of both orders must match to prevent a trade. If specified on both orders, Trading Group ID (character 3) must match to prevent a trade. Note that in the event of a Spread order match with a Simple order, the Spread order will always be cancelled irrespective of the value of the 1st character.
Price	8	Binary Price	Corresponds to *Price* (44) in CFE FIX. Limit price. Four implied decimal places. Required for limit orders (*OrdType* = 2). If specified on market order (*OrdType* = 1), the order will be rejected. Orders will be rejected if Price does not fall on the applicable minimum trading increment. For all contracts other than VXT, simple orders will be rejected if *Price* is less than or equal to zero, or greater than or equal to 100,000. For VXT, simple orders will be rejected if *Price* is outside the price limits presented in the contract specification. Spread orders will be rejected if *Price* is outside the price limits implied by the spread instrument definition and constituent instrument min and max prices.
ProductName	6	Text	Used to specify product class (e.g., "VX", "VU, etc.) for `Purge Orders` and `Cancel Order` message cancel by product functionality. If an unrecognized ProductName is provided, the associated request will be rejected with reason code C (Unknown Product Name).

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RiskReset	8	Text	Corresponds to *RiskReset* (7692) in CFE FIX. *Single Character Values (Values may be combined)* S = Product-level risk/lockout reset F = Firm-level lockout reset C = CustomGroupID lockout reset Values may be combined together to allow for resets of multiple risk trips or self-imposed lockouts in a single message. For example, "FS", "SC", "FC", and "SFC" are all acceptable values. The characters may be combined in any order. For example, to "reset all" set this field to "SFC", which is the equivalent to "CFS". For more information, refer to the CFE US Futures Risk Management Specification.
SecondaryExecID	8	Binary	Corresponds to *SecondaryExecID* (527) in CFE FIX. Field indicates whether an execution is a Spread instrument execution or a Simple instrument execution that is part of a Spread execution. • If *SecondaryExecID* field is not present, the execution is a Simple instrument execution only. • If *SecondaryExecID* is present and is the same as the *ExecID* required field, the execution represents a Spread execution for which associated Simple instrument executions will follow. • Simple instrument executions associated with a Spread execution will contain a *SecondaryExecID* value that matches the *ExecID* of the associated Spread execution.
SecondaryOrderID	8	Binary	Corresponds to *SecondaryOrderID* (198) in CFE FIX. For MTP triggered `Order Cancelled` message, value contains the *OrderID* of the other order in the MTP pair.
Side	1	Alphanumeric	Corresponds to *Side* (54) in CFE FIX. 1 = Buy 2 = Sell
StopPx	8	Binary Price	Corresponds to *StopPx* (99) in CFE FIX. Stop price. Required if *OrdType* = 4 (Stop Limit). Stop Limit orders will only be triggered off Last Sale Eligible trades.
SubLiquidityIndicator	1	Alphanumeric	Additional information about the liquidity of an order. CFE may add additional values without notice. TPHs must gracefully ignore unknown values. ASCII NULL (0x00) = No Additional Information C = Carried Order Indicator U = Qualifying Market Turner order

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Symbol	8	Alphanumeric	Corresponds to *Symbol* (55) in CFE FIX. Simple Instruments can be specified by providing the mapped symbol format in the Symbol field or by providing the product name (e.g., "VX") in the *Symbol* field and maturity date in the *MaturityDate* field. Responses to the TPH will contain the instrument specification in the manner that was provided on the associated new order specification (e.g., either Symbo ID or Product and MaturityDate). The Symbol field for Spread instrument related messages will always contain mapped symbol ID as product and maturity date does not completely specify the Spread instrument.
TimeInForce	1	Alphanumeric	Corresponds to *TimeInForce* (59) in CFE FIX. `0` = Day (Expires at the end of the business day.) `1` = GTC (Good 'till Cancel. Order remains until cancelled or contract expires). `3` = IOC (Portion not filled immediately is cancelled. Market orders are implicitly IOC.) `4` = FOK (An IOC where the entire size must be filled, else the order will be cancelled back) `6` = GTD (Good 'till Date-Time Expires at the date-time specified in the *ExpireTime* field).
TradeDate	4	Date	Corresponds to *TradeDate* (75) in CFE FIX. TradeDate represented as Note that on CFE, business date is not always the same as the calendar date. For example, the VX/VT products open for trading on the calendar day prior to the associated business date. Executions that occur after the open and before midnight will have a *TradeDate* value that is not the same as the calendar date of the execution.

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8 Reason Codes

8.1 Order Reason Codes

The following is a list of all reason codes used by CFE. These reason codes are used in a variety of contexts (order cancellations and order rejections). All reasons are not valid in all contexts. The reason code will be followed by free-form text. The specific text the system delivers may vary from the test listed below, to provide clarification of the reject reason. CFE may add additional reason codes without notice. Members must gracefully ignore unknown values.

`A` = Admin

`B` = Unknown maturity date

`C` = Unknown product name

`D` = Duplicate identifier (e.g., ClOrdID)

`H` = Halted

`I` = Incorrect data center

`J` = Too late to cancel

`K` = Order rate threshold exceeded

`M` = Liquidity available exceeds order size

`N` = Ran out of liquidity to execute against

`O` = *ClOrdID* doesn't match a known order

`P` = can't modify an order that is pending

`U` = User requested

`V` = Would wash

`X` = Order expired

`Y` = Symbol not supported

`Z` = Unforeseen reason

`h` = Order persisted

`f` = Risk management EFID level or custom group ID level

`m` = Market access risk limit exceeded

`n` = Risk management configuration is insufficient

`o` = Max open orders count exceeded

`s` = Risk management product level

`y` = Order received by CFE during replay

`z` = Session end

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8.2 Quote Reason Codes

The following is a list of all quote reason codes used by CFE. All reasons are not valid in all contexts. The reason code will be followed by free-form text. The specific text the system delivers may vary from the test listed below, to provide clarification of the reject reason. CFE may add additional reason codes without notice. Members must gracefully ignore unknown values.

`C` = Invalid EFID (*ClearingFirm*)

`D` = Invalid *WashId*

`F` = Not enabled for quotes

`I` = Incorrect data center

`K` = Message rate threshold

`L` = Invalid *QuoteCnt*

`M` = Symbols not on same matching engine

`O` = Invalid ManualOrderIndicator

`Q` = Invalid *QuoteUpdateID*

`R` = Futures root does not match across quotes

`S` = Symbol not found

`W` = Invalid *WashPreventType*

`a` = Admin

`c` = Invalid *Capacity*

`e` = Invalid OEOID (Order Entry Operator Id)

`f` = Risk management EFID or Custom Group ID level

`i` = Invalid CtiCode

`m` = Invalid *WashMethod*

`n` = Exceedes max notional value per order

`o` = Invalid Open/Close

`p` = Risk management product level

`r` = Invalid Remove

`s` = Invalid *Side*

`u` = Symbol range unreachable

`x` = Exceedes max size per order

`y` = Quote received by CFE during replay

`z` = Invalid *SizeModifier*

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9 List of Message Types

9.1 TPH to CFE

Message Name	Level	Type	Sequenced
Login Request	Session	0x37	No
Logout Request	Session	0x02	No
Client Heartbeat	Session	0x03	No
New Order	Application	0x38	Yes
Cancel Order	Application	0x39	Yes
Modify Order	Application	0x3A	Yes
Quote Update	Application	0x70	Yes
Purge Orders	Application	0x47	Yes
Reset Risk	Application	0x56	Yes

9.2 CFE to TPH

Message Name	Level	Type	Sequenced
Login Response	Session	0x24	No
Logout	Session	0x08	No
Server Heartbeat	Session	0x09	No
Replay Complete	Session	0x13	No
Order Acknowledgment	Application	0x25	Yes
Order Rejected	Application	0x26	No
Order Modified	Application	0x27	Yes
User Modify Rejected	Application	0x29	No
Order Cancelled	Application	0x2A	Yes
Cancel Rejected	Application	0x2B	No
Order Execution	Application	0x2C	Yes
Trade Cancel or Correct	Application	0x2D	Yes
Purge Rejected	Application	0x48	No
Mass Cancel Acknowledgment	Application	0x36	No
TAS Restatement	Application	0x49	Yes
Variance Restatement	Application	0x4A	Yes
Quote Update Acknowledgment	Application	0x71	No
Quote Restated	Application	0x72	Yes
Quote Cancelled	Application	0x73	No
Quote Execution	Application	0x74	Yes
Reset Risk Acknowledgment	Application	0x57	No
Quote Update Rejected	Application	0x78	No
TAS Quote Restatement	Application	0x75	Yes
Variance Quote Restatement	Application	0x76	Yes

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10 Port Attributes

The table below lists BOE port attributes that are configurable on the port or firm level. Changes to these attributes can be made by contacting the CFE Trade Desk.

Attribute	Default	Description
Allowed Executing Firm ID(s)	All EFIDs	Executing Firm ID(s) allowed for trading on the port.
Default Executing Firm ID	None	Default Executing Firm ID to use if none is sent on a `New Order`.
Cancel on Disconnect	All	Cancels open orders upon order handler disconnect; both graceful and ungraceful. If Cancel On Disconnect is set, open orders in products that are not in Closed state at the time of the disconnect are cancelled. `All` = Cancel Day, GTC, and GTD orders Day = Cancel only Day orders None = Disabled BOE Quoting ports require Cancel on Disconnect set to All or Day. Default will be used if not specified.
Cancel on Reject[1, 3]	No	Cancels an order upon a cancel or modify reject for that order.
Cancel on ME Disconnect	All	Controls whether orders are cancelled or preserved on a Matching Unit failover and provides for the ability to preserve GTC orders (Day). In any event, if a failover takes longer than 5 minutes, all orders are cancelled (including GTCs). `All` = Cancel Day, GTC, and GTD orders Day = Cancel only Day orders None = Disabled BOE Quoting ports require Cancel on Disconnect set to All or Day. Default will be used if not specified.
Cancel Open Orders on DROP Port Disconnect	No	Only applicable if "Reject Orders on DROP Port Disconnect" has been enabled. When the last Standard FIX DROP port associated with an order handler session has disconnected, open orders, associated with the session are cancelled. `All` = Cancel Day, GTC, and GTD orders Day = Cancel only Day orders None = Disabled Note this parameter applies to Standard FIX DROP ports and not Order-By-Order DROP ports (ODROP).
Carried Order Restatements	Yes	If the Carried Order Restatements port attribute is set, `Order Acknowledgement` messages representing orders carried forward from the previous business date will be sent after the `Login Response` message and before regular session messages for each product. See 'Section 1.5.1 - Carried Order Restatements' for a detailed description of Carried Order Restatements. Note that any changes made to any port attribute will not be enforced on carried GTC orders. Members who wish to apply updated port attributes to resting GTC orders must cancel those orders, and then resubmit

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		them following the effective time of the port attribute change.
Default MTP Value[1]	None	Specifies default value for *PreventMatch*.
Default Customer Order Handling Instruction	Y = Electronic	Sets a default CustOrderHandlingInst (1031) that will be used unless overridden at the individual order level. W = Desk (high touch) Y = Electronic (default) C = Vendor-provided platform, billed by Executing Broker G = Sponsored Access via Exchange API or FIX, provided by executing broker H = Premium algorithmic trading provider, billed by executing broker D = Other, including other-provided screen
Maximum Order Size	25,000 contracts	A system-wide maximum order size limit that is set by the CFE. TPHs may not request a change to this port attribute.
Reject Orders on DROP Port Disconnect	No	Allows TPH/Sponsoring Firms to associate DROP port(s) to order entry port(s). If all associated DROP ports experience disconnection, new orders will be rejected until at least one DROP port session has been reestablished.
Reject Orders on DROP Port Timeout(s)	30 seconds	Only applicable if "Reject Orders on DROP Port Disconnect" has been enabled. When the last associated DROP port has disconnected, begin rejecting orders on the associated order entry port(s) if a DROP session has not been reestablished within this timeout. Minimum value allowed is 0 seconds.
Send Trade Breaks[2]	No	Enables sending of `Trade Cancel` or `Correct` messages.
Port Message Rate Threshold	Default and Max allowed = 3,000 msgs/sec for order ports,10,000 for quote ports. 10 msg/sec for CFE test products.	The maximum allowed message rate on the session. When the first non-administrative message is received, a one second window begins. For example, on an order port during a second no more than 2,999 additional non-administrative messages will be allowed within that window. If the rate is exceeded all new orders in the time window are rejected, modifies are treated as cancels, and cancels are processed. Quote ports will have a default limit of 10,000 messages per second. A message is defined as any individual quote or any order-related message (new, modify, or cancel). If the limit is exceeded on a quote port the behavior for orders described above applies, and all `Quote Update` messages will be rejected unless the `Quote Update` contains only cancels. Note: Order handler burst rates towards each matching unit may be limited as described in 'Section 1.5.1 – Architecture'.

[1] Port attribute can be overridden on an order-by-order basis

[2] Requires certification

[3] Not applicable for quotes

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Symbol Message Rate Threshold	Default and Max allowed = 3,000 msgs/sec for order ports and 10,000 for quote ports.	Functions the same as the Port Message Rate Threshold but is calculated at the symbol level. It is capped by the Port Message Rate Threshold. Note: Order handler burst rates towards each matching unit may be limited as described in 'Section 1.5.1 – Architecture'.

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11 Support

Please email questions or comments regarding this specification to cfetradedesk@cboe.com.

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Revision History

Version	Date	Description
1.0.0	05/01/17	Initial version.
1.0.1	05/17/17	Added Cancel on ME Disconnect port attribute. Changed *Binary Price* to a signed data type. Removed *Signed Binary Price*, *Signed Binary Fee*, and *Short Binary Price* data types. Cleanup of minor typos and formatting.
1.1.0	07/14/17	Added additional fields to accommodate Spread instrument executions. Updated Mass Cancel and Purge fields to add additional filtering based on GTC orders and Spread instruments.
1.1.1	07/24/17	Introduced improved method of specifying Mass Cancel and Purge Orders operations using the MassCancelInst field. Modified and clarified explanation for Variance and TAS pending and restatement Execution Reports and associated custom fields for transformed symbol, price and size for clearing. Modified TAS and Variance Restatement messages to contain same required fields as `Order Executed`.
1.1.2	08/07/17	Renamed *VariancePrice*, *VarianceSize* and *NewSymbol* to *ClearingPrice*, *ClearingSize* and *ClearingSymbol* respectively. Removed *TASPrice* field using *ClearingPrice* instead to report transformed TAS execution price. Removed *BaseLiquidityIndicator* from `Order Execution` message list of valid optional fields.
1.1.3	08/14/17	Corrected `Purge Orders` input bitfields specification.
1.1.4	08/25/17	Clarified ClOrdID uniqueness in New Order to account for long-lived GTC orders. Added N value to optional field OpenClose. Moved general presentation of protocol features to Section 1 – Introduction. Clarified that Cancel on Disconnect applies for graceful and ungraceful disconnect. Updated Cancel on ME Disconnect and Cancel on DROP Port Disconnect port attributes to provide ability to filter out GTC orders.
1.1.5	09/12/17	Updated description of Carried Order Restatements section to clarity steps required to receive on connect including specifically enabling replay for associated matching units in the `Login Request` message.
1.1.6	09/25/17	Added to documentation of *Price* field of `New Order` message to specify that orders that don't comply with tick increments or values outside of system price limits will be rejected. Update Return Bitfields for `Order Modified` message; *OrigClOrdId, LeavesQty, LastShares, LastPx and BaseLiquidityIndicator* are value return fields. Updated explanatory text for reason code 'n' (risk). Added explanatory text for *MassCancelInst* lockout behavior. Updated bitfields in return messages, including `Order Modified` *LeavesQty*.

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1.1.7	10/11/17	Removed *EchoText* optional field from `New Order` input message and removed from various return messages. Removed *LastShares* and *LastPx* optional return bitfields from `Order Modified` message. Removed *LastShares, LastPx* and *BaseLiquidityIndicator* optional return bitfields from `Order Cancelled` message. Removed all Byte 12 return bitfields from `Trade Cancel or Correct` message. Added Byte 5 return bitfields (*OrigClOrdId, LeavesQty, LastShares and LastPx*) and removed Byte 13 return bitfields (*CumQty* and *AvgPx*) from `TAS Restatement` message. Added Byte 5 return bitfields (*OrigClOrdId, LeavesQty, LastShares* and *LastPx*) and removed Byte 13 return bitfields (*CumQty, DayOrderQty, DayCumQty, AvgPx,* and *DayAvgPx*) from `Variance Restatement` message.
1.1.8	10/17/17	Cboe branding/logo changes.
1.1.9	10/30/17	Clarified Account field is a unique identifier associated with an order. Added calculation of trade size in Variance units to VA pending `Order Execution` message *ClearingSize* field. Added associated return bitfield for `Order Execution` message. For consistency with VA, added *ClearingSize* to pending `Order Execution` messages for VAO trades.
1.1.10	11/02/17	Clarified definition of *BaseLiquidityIndicator* field, associated with FIX *TradeLiquidityIndicator* (9730) and added "C" value for market-opening/reopening. Removed header fields *LastShares*, *LastPx*, *LeavesQty*, *BaseLiquidityIndicator*, *SubLiquidityIndicator* and *ContraBroker* from `TAS Restatement` and `Variance Restatement` messages. Removed *OrderQty*, *LeavesQty*, *BaseLiquidityIndicator* and *PendingStatus* fields from `TAS Restatement` and `Variance Restatement` return bitfields.
1.1.11	11/14/17	Fixed ordering of fields in `New Order` message example. Added Port Order Rate Threshold and Symbol Order Rate Threshold port attributes. Updated description of *MinQty* field in `New Order` message to clarify usage. Added support for GTD orders comprising addition *TimeInForce* value "6", addition of *ExpireTime* to `New Order` message and *ExpireTime* as valid return field for `Order Acknowledgement`, `Order Modified`, `Order Cancelled`, `Cancel Rejected` and `Order Execution` messages. Effective 12/11/17.
1.1.12	11/22/17	Removed 'Report MTP Fields' port attribute, which is applicable only for FIX. Added *LastPx* and *LastShares* to return fields for `Order Cancelled` and updated description of each field to indicate they are present on MTP triggered `Order Cancelled` messages.
1.1.13	12/05/17	Combined the two 'f' Reason Codes into one line for clarity.

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1.1.14	12/21/17	Clarified use of Expiration symbology vs. mapped Symbol ID for Spread execution Execution Reports. Specifically, updated description of *Symbol* and *MaturityDate* fields. Added *OperatorId* and *ManualOrderIndicator* as fields that may be changed with a `Modify Order` message. Clarified that *OperatorId* and *ManualOrderIndicator* are required on all TPH to CFE message types. Added GTC and GTD *TimeInForce* values as accepted for Stop and Stop Limit orders.
1.1.15	01/19/18	Corrected `New Order` message example to reflect that *Account* is a required field. Updated description of the *Capacity* (47) to clarify meaning of C and F with respect to OCC ranges.
1.1.16	02/27/18	Minor corrections to example messages. Added session availability times and description of daily restart to Hours of Operation section. Documented the system limit of 1,295 `Modify Order` requests per order per day.
1.1.17	03/05/18	Added OCC Clearing Reference section to more clearly describe the BOE to OCC field mappings. Maximum number of `Modify Order` requests per order per day will be raised to 1,679,615 effective 3/18/18.
1.1.18	03/14/18	Clarified description of the case where overlapping `Modify Order` messages may be used. Clarified valid Account field characters are ASCII 32-126 (i.e., space is a valid character). Effective 3/18/18, updated definition of the OperatorId field to include characters 33-126, except for comma, semicolon, and pipe with minimum length of 3 and maximum length of 18 characters.
1.1.19	04/10/18	*CumQty* to be populated on leg fills related to complex executions (Effective 4/29/18).
1.2.0	04/26/18	Added optional fields to the `Purge Rejected` message to accommodate optional return of the *MassCancelId* field from the associated `Purge Request` message (Effective 7/1/18). Updated *OperatorID* field name to *OEOID*. Added section to Protocol Features detailing the conditions under which persisted orders can be cancelled while the associated product is in a suspended state.
1.2.1	05/30/18	*MassCancelId* moved to bit 8 from bit 1 in byte 15 of the Return Bitfields for a `Purge Rejected` message. Added additional detail around holiday schedules and when order cancellations are processed on holidays.
1.2.2	07/02/18	Correction to example for `Purge Rejected` message.
1.2.3	07/20/18	Added *SubLiquidityIndicator* value of "U" to indicate Market Turner status.
1.2.4	08/07/18	Added notes to Cancel Order and Purge Orders sections indicating request limits of 20 per second for duplicative requests (effective 8/15/18).

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1.2.5	10/09/18	Added *FrequentTraderId* optional field to support future billing related changes (effective 11/1/18).
1.2.6	10/10/18	Added description of Maximum Order Size limit set by CFE in Port Attributes section.
1.2.7	11/14/18	Correction to Return Bitfield information for `Cancel Rejected` message to indicated *Frequent TraderID* cannot be requested for the `Cancel Rejected` message.
1.2.8	02/27/19	Corrected both examples for `Purge Orders` message.
1.2.9	06/12/19	Removed reference to Firm Risk Reset as they do not apply to CFE.
1.2.10	07/02/19	Corrected *Symbol* field length from 6 to 8 in the `New Order` message example.
1.2.11	08/07/19	Changed Return Bitfield *EquityNBBOProtect* to "Reserved".
1.3.0	12/20/19	Added description of BOE Quote Interface and Reset Risk functionality (effective Q2, 2020)
1.3.1	12/24/19	Updates to BOE Quoting Interface message examples. Clarified field descriptions of `Reset Risk` message. Updated descriptions of *RiskReset*, and *MassCancelInst* to accommodate new risk reset functionality and addition of quotes.
1.3.2	02/19/20	Added values of e, i, O, to Quote Reason Codes in support of BOE Quote Interface. Added note indicating that the reason code text the system delivers may vary from the test listed, to provide clarification of the reject reason. Updated return bitfield tables. Added `Quote Restated`, `TAS Quote Restatement`, `Variance Quote Restatement` message sections (effective with BOE Quoting Interface).
1.3.3	03/05/20	Added *SizeModifier* field to `Quote Update` message to allow TPHs optionality around handling size changes to quotes. Added message rate limits of 10,000 messages per second for quote ports.
1.3.4	04/06/20	Updated Quotes Reason Codes for "h", "n" and "x". Invalid CTI Code quote reason code was updated from "I" to "i". Added "z = Session End" and "h=order persisted" Order Reason Codes. Updated BOE Quote Interface and Reset Risk effective date to 5/3/20.
1.3.5	04/21/20	Updated formatting. Updated MassCancelID requirement for combination of values "M","S" and/or "B". Updated BOE Quoting Ports launch date. Effective date to 5/3/20 for TAS contracts and 7/26/20 for all other futures contracts.
1.3.6	05/05/20	Added Maximum Order Limit section. Updated reference to test product ZVXT entering a queuing state 5 minutes prior to other CFE products. Added message rate limits of 10 messages per second for CFE Test Products (effective TBD).
1.3.7	05/07/20	Postponement for changes to ZVXT trading schedule and test message rate limit.

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1.3.8	05/13/20	Added information regarding test product ZVXT entering a queuing state 5 minutes prior to other CFE products. Updated message rate limits to 10 messages per second for CFE test products (effective 5/31/20).
1.3.9	05/21/20	Added *CustOrderHandlingInst* field to `New Order` and `Modify Order` messages as well as an optional return field on several return messages. The field may be defaulted at the port level using the 'Default Customer Order Handling Instruction' port attribute (effective 6/28/20).
1.3.10	09/22/20	A change to *Capacity* for a quote must be done by cancelling the original quote and sending a new quote. Added note indicating VXT end-of-day settlement price will be available after 3:00 p.m. CT (effective 10/26/20).
1.3.11	12/11/20	Added detail to *ClOrdID* description to indicate that a leading tilde (~) cannot be sent on any *ClOrdId* and will result in a reject.
1.3.12	03/11/21	Added Section 1.5.1 – Architecture to provide high level overview of protocol architecture and source IP blocking feature. Added 'n' rejection value in *QuoteResult* field.

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